Bye Bye Beta:
Deposit Duration with Fixed Spreads

Robert Rogers*

First version: November 2024

This version: October 2026

Click here for the latest version

*University of Chicago Booth School of Business. Email: robert.rogers@chicagobooth.edu. I thank Dimitri Vayanos, Martin Oehmke, Ian Martin, Peter DeMarzo, Arvind Krishnamurthy, Christian Julliard, and seminar participants at the London School of Economics and the SITE 2026 conference on banking and the deposit franchise at Stanford University for helpful comments and discussions. This paper forms part of my dissertation at the London School of Economics.

Bye Bye Beta:
Deposit Duration with Fixed Spreads

Abstract

Are very low interest rates bad for bank value? Under the standard fixed "deposit beta," they can be only through fixed costs or shrinking deposits. I show that deposit rates instead behave like a fixed spread below market rates, floored at zero: the payoff of an interest rate option. Replicating deposits with traded option prices, I find deposit spreads themselves lose value steeply as long-term rates approach zero. The resulting measure of bank interest rate risk predicts stock responses to monetary policy shocks, where existing measures fail. Banks partly hedged this risk in 2020–21. Fully hedging it would have required far larger long-term securities purchases, increasing losses and run risk once rates rose.

Keywords: duration, interest rate risk, deposit pricing, deposit spreads, franchise value

JEL classification: G12, G21

Are very low interest rates bad for banks? Banks and analysts seemed to think so—a typical equity analyst report from early 2022 lists "rates normalization" as a "reason for hope" for bank stocks (Morgan Stanley & Oliver Wyman, 2022).[1] And at Silicon Valley Bank, a focus on "protection from potential rate decreases" led management to remove the hedges on its securities portfolio in 2022 (Federal Reserve Board, 2023).

But a straightforward look at the income statements of banks makes it hard to see why. DeMarzo et al. (2024) decompose bank income into a fixed positive stream and a floating positive stream. Fixed income values increase when rates fall, and floating income values remain constant. So the value of the bank should always benefit from lower rates. Were banks and analysts simply mistaken? Should regulators be assuming that interest rate cuts always increase bank equity value?

This paper shows that the value of deposit spreads and the value of the bank can indeed decline sharply at low rates. This decline emerges naturally once we drop the assumption that deposit rates are a fixed fraction (a fixed "beta") of the risk-free rate, and it occurs even before considering fixed costs or deposit runoff. Measuring this value with a replicating portfolio of interest rate options, I find that it drives large variation in bank interest rate sensitivity at low rates, and predicts the response of bank stock prices to monetary policy shocks.[2] This time-varying sensitivity helps explain why banks bought so many long-term securities in 2020–21: the securities hedged the interest rate sensitivity of bank equity.

The mechanism can be illustrated by a simple example with a permanent depositor in a bank. Suppose the interest rate received by the depositor is some function of the risk-free rate, $f(r_t^f)$, and the risk-free rate is fixed at some low rate ϵ with certainty until time T. If we denote the value at time T of all future interest payments after time T by FV_T, then

[1] DeMarzo, Krishnamurthy, and Nagel (2024) also document that bank 10-K disclosures state that higher rates will increase equity values.

[2] Throughout, "duration" refers to the percentage fall in value from a one percentage point rise in interest rates, and "interest rate sensitivity" to its negative, i.e., the percentage rise in value. Positive interest rate sensitivity therefore means negative duration.

the value of the deposit interest payments is:

$$PV_0 = \underbrace{\frac{f(\epsilon)}{\epsilon} \times (1 - e^{-\epsilon T})}_{\text{Value up to time T}} + \underbrace{FV_T \times e^{-\epsilon T}}_{\text{Value after time T}}$$

The effect of a small increase in the near-term rate ϵ when ϵ is low will be:

$$\lim_{\epsilon \to 0} \frac{\partial PV_0}{\partial \epsilon} = \underbrace{T \times f'(0)}_{\text{Cash flow effect}} - \underbrace{T \times FV_T}_{\text{Discounting effect}}$$

If deposit interest rates are exactly linear, as in the standard "deposit beta" model in which $f(r) = \beta \times r$, then $f'(0) = FV_T$, so this derivative is zero and the value of deposits is always β (e.g., Drechsler, Savov, and Schnabl, 2021; DeMarzo et al., 2024).

However, the banking literature has shown that the "pass-through" of interest rates to deposit rates, $f'(r_t^f)$, varies depending on the interest rate (e.g., Kang-Landsberg, Luck, and Plosser, 2023; Greenwald, Schulhofer-Wohl, and Younger, 2023; Xu, 2024; Lu and Wu, 2025). In particular, pass-through seems to be near zero at very low rates, as deposit rates are squeezed up against the lower bound, i.e., $f'(0) \approx 0$. In this case, the value of the derivative will be negative. While deposit rates are pinned at zero, a rise in rates does not raise deposit payments, but it does lower the present value of the payments that resume once rates normalize, increasing the value of the bank.

For illustration, suppose rates are held at 0 for 5 years ($T = 5$), pass-through is 0 at the zero lower bound ($f'(0) = 0$), and the value at time T of interest payments on a dollar of deposits is 80 cents ($FV_T = 0.8$). Then $\frac{\partial PV_0}{\partial \epsilon} = -4$, and a 1 ppt increase in the interest rate will lower the value of bank deposit liabilities by 4 cents per dollar of deposits. If the bank is 10× leveraged, this translates to a 40% increase in the value of equity.

To show that this decline in the value of deposit spreads at low rates is quantitatively important in reality, I measure time-varying deposit value and duration with a replicating portfolio approach. First, I estimate the relationship between the risk-free rate and aggre-

gate interest income on interest-bearing instant-access deposits. I fit a flexible econometric model and find that the long-term pass-through of risk-free rate changes varies considerably depending on the interest rate level. At low rates there is very little pass-through, and rates are nearly pinned at 0. At high rates, the long-term pass-through approaches one, and the rate is well-approximated by a "fixed spread" of two percentage points ($f(r_t^f) \approx r_t^f - 2\%$), instead of a fixed beta, although short-term pass-through remains low.

I then value a permanent dollar of these deposits by replicating this deposit rate function with a portfolio of call options on the interest rate, known as interest rate "caps." Each quarter, a cap with strike K pays $\max\{r_t^f - K,\, 0\}$. Combining caps across a range of strikes replicates the estimated deposit rate function (as in Breeden and Litzenberger, 1978), so their prices give a measure of deposit value that does not depend on a model of interest rates. The only assumptions needed are those used to fill in unquoted maturities and strikes.

Figure 1 plots this replication-based estimate of deposit franchise value (i.e., one minus the present value of deposit payments) against the 10-year zero-coupon rate. Franchise value declines at both high and low rates, and particularly steeply at low rates. At low rates, deposit interest is pinned at zero and a rise in rates increases bank value. At high rates, deposit spreads behave like an ordinary fixed-income stream and lose value as rates rise. If deposit interest rates instead followed a constant "beta," this hump shape would disappear and the plot would be a flat horizontal line.

The duration of the bank's assets does not drown out this variation in interest rate sensitivity. To show this, I conduct a full valuation of the aggregate banking industry, including all assets and fixed costs, using balance sheet data from the FFIEC Call Reports and a model of interest rates. Deposit spreads contribute substantially more variation to bank duration than assets do, although the level of total bank duration depends on assumptions about fixed costs.

I validate this bank duration estimate using high-frequency monetary policy shocks around FOMC announcements. Under my baseline measure, predictions match observed

bank stock price responses with the correct sign, magnitude, and high statistical significance ($R^2 = 10\%$ for announcement-day returns, excluding the financial crisis), as shown in Figure 2. In contrast, existing balance-sheet-based measures of bank duration—including measures constructed based on DeMarzo et al. (2024), Begenau, Piazzesi, and Schneider (2015), and Haddad and Sraer (2020)—either fail to predict returns or predict them with the wrong sign. Much of this predictive power is driven by the estimated deposit duration, although the duration of modeled fixed costs also plays an important role.

These estimates also raise questions about how bank interest rate risk should be measured. Traditionally, bank asset-liability management has focused on the sensitivity of net interest income over short and medium horizons. Recent academic work instead tends to treat interest rate risk as a question of equity value and duration (e.g., DeMarzo et al., 2024; He, Krishnamurthy, and Milbradt, 2026), and the Basel Committee's standards on interest rate risk in the banking book ask banks to measure the effect of rate shocks on their "economic value of equity" (BCBS, 2016).[3] The business press has likewise often attributed the 2020–21 build-up of long-term securities to a myopic focus on short-term interest income rather than value (e.g., Doherty, 2023; Wilkes, 2023; Peters, 2023; Nichols, 2023).

My estimates suggest that truly hedging value and neutralizing duration would not have prevented this build-up. Neutralizing the duration of bank equity would have required banks to buy even more long-term securities during the low-rate period, and the 2020–21 purchases can be seen as a partial attempt to do so. While these long-term securities reduced the volatility of bank equity when they were purchased, they left banks with greater shortfalls and potential losses in the event of a bank run, as in Drechsler, Savov, Schnabl, and Wang (2023) and Haddad, Hartman-Glaser, and Muir (2023). And because deposit franchise value is hump-shaped in rates, the hedge reversed once rates rose: the same securities added to bank interest rate risk from 2022 onward.

[3]The specifics of the Basel standards on EVE calculations differ from the methodology of this paper, and I do not mean to imply that Basel guidance and principles related to modeling of EVE caused banks to buy more long-term securities.

Related literature

Several related papers consider the effects of the interest rate level on bank and deposit franchise duration. DeMarzo et al. (2024) argue that bank value always benefits from lower interest rates. They frame the puzzle addressed by this paper: if deposit spreads are approximately a floating rate and quantities are growing, then why would value decline at low rates?

The key contribution of this paper is to show that deposit spreads alone can decline in value at low rates, even without any fixed costs or deposit outflows. Greenwald et al. (2023); Drechsler et al. (2023); Koont, Santos, and Zingales (2025); Lu and Wu (2025) and other papers propose that low interest rates can hurt bank valuations, as argued in this paper. But they ultimately rely on either negative fixed cash flows or shrinking deposits to deliver the negative duration (as argued in DeMarzo et al., 2024). Greenwald et al. (2023) and Koont et al. (2025) specifically consider the effects of varying deposit pass-through rates on duration. Because their valuation framework assumes a constant risk-free rate, their formulas deliver positive duration if deposit runoff and fixed costs are set to zero. English, Van den Heuvel, and Zakrajček (2018) measures interest rate sensitivity through the effect of monetary policy shocks on bank stock prices, but focuses on the pre-crisis period and so finds positive duration.

The empirical pass-through estimation in Section 1 draws on the existing empirical literature on deposit betas. As in Kang-Landsberg et al. (2023); Wang (2020); Di Tella and Kurlat (2021); Greenwald et al. (2023); Xu (2024); Lu and Wu (2025), I find that pass-through is higher when rates are higher. My average estimates of short-term pass-through are broadly similar to the estimates of pass-through from the one-year beta regressions in Drechsler et al. (2021). I find more non-linearity in pass-through rates than DeMarzo et al. (2024), and I include a reconciliation of the source of our differences in Section 1 and Appendix B.

Begenau et al. (2015) and Haddad and Sraer (2020) also create time-varying measures of aggregate bank duration based on bank balance sheet data. They find that these measures

do not predict the relationship between interest rate changes and stock returns well. Haddad and Sraer (2020) frame this disconnect as a puzzle. This paper offers an explanation: measures focused on assets and statutory liabilities miss the large, time-varying deposit franchise duration.

Finally, the paper relates closely to Drechsler et al. (2023) and Haddad et al. (2023), who explain the 2023 regional banking crisis through the deposit franchise. These papers propose that the deposit franchise has negative duration because of deposit runoff and fixed costs, but that its value disappears if depositors run. Hence if banks try to hedge this franchise duration by buying long-term securities, they risk runs when rates rise. This paper shows that the hedging motive behind this dilemma arises even without any runoff or operating costs, and that because deposit rates are floored at zero, it was particularly strong when rates were very low.

Contents

Section 1 presents the empirical model of deposit rates. Section 2 calculates the replicating-portfolio value of deposits. Section 3 estimates whole-bank duration and validates it against monetary policy shocks. Section 4 examines the implications for bank hedging and long-term securities purchases. Section 5 concludes.

1 Deposit spreads

To value deposits, I first estimate the relationship between deposit rates and the risk-free rate. I find that long-term deposit spreads are approximately fixed when interest rates are high, and compare my results to other empirical estimates of the deposit interest rate function.

1.1 Model

The banking literature provides two different benchmarks for the relationship between the risk-free rate and deposit rates. Much recent empirical work on banking models deposit

rates with a fixed "beta": $f(r_t^f) = \beta r_t^f$ (e.g., Drechsler et al., 2021; DeMarzo et al., 2024). On the other hand, discrete-choice models used in the industrial organization literature tend to deliver deposit spreads that are roughly constant with respect to the risk-free rate, with deposit rates floored at 0: $f(r_t^f) \approx \max\{r_t^f - \delta, 0\}$. This is true for both simple logit-based exercises (e.g., Dick, 2008) and more thorough modeling of the competition between deposits and cash (Wang, Whited, Wu, and Xiao, 2022).[4]

To distinguish between these two possibilities, I fit a general model that allows for deposit spreads to be fixed or proportional. More specifically, in this section I specify a simple time-series model of deposits that allows for (a) either fixed or proportional spreads, (b) a 0 floor on deposit interest rates, and (c) a delay in deposit pass-through. In Appendix A I show that the empirical model can be derived from a discrete-choice model of bank competition with a zero lower bound and a lagging customer perception of the risk-free rate.

The model posits that an unobserved "shadow" deposit rate (r_t^d) follows a simple AR(1) relationship with the short-term risk-free rate:

$$r_t^d = \gamma r_{t-1}^d + \beta(1-\gamma)(r_t^f - \delta) + \varepsilon_t \tag{1}$$

The observed deposit rate (r_t^o) follows a non-linear censoring function of this shadow rate such that the observed rate is never negative. The censoring function is based on the idea that different depositors receive different rates, with cross-sectional variance s^2, and no depositor can be paid less than 0:

$$r_t^o = g(r_t^d) \equiv r_t^d \underbrace{\Phi\left(\frac{r_t^d}{s}\right)}_{\text{Share paid} > 0} + \underbrace{s\phi\left(\frac{r_t^d}{s}\right)}_{\text{Adjustment from censoring lowest-paid depositors}} \tag{2}$$

where Φ denotes the standard normal CDF and ϕ the standard normal PDF. The motivation and parameters of this model are described below.

[4]For example, Figure 4 of Wang et al. (2022) shows that modeled pass-through is approximately 1 when the risk-free rate is over 2%.

Equation (1) is a simple AR(1) with one exogenous variable. I have simply expressed the constant parameter as $-\beta(1-\gamma)\delta$ and the coefficient of the exogenous variable as $\beta(1-\gamma)$ to give the variables intuitive interpretations. The parameter δ represents the long-term fixed spread between deposit rates and the risk-free rate, as we would expect in a simple discrete-choice model. The parameter β represents the long-term proportional spread as in Drechsler et al. (2021). And γ allows for delays in deposit pass-through, which have been empirically well documented since at least Hannan and Berger (1991).

The importance of allowing for this lag in pass-through can be seen from the time series of deposit rates, shown in Figure 3. Long-term pass-through is clearly higher than short-term pass-through—for example, deposit rates showed a small response to the sudden drop in risk-free rates in 2001, but seem to have mostly followed the long decline from 1990 to 2010. Neglecting this delay in pass-through would force the model to fit the same pass-through rate for long and short periods, underestimating long-term pass-through.

If $\delta > 0$, this model would lead to negative deposit rates when the risk-free rate is low. Since in reality we do not observe any significant negative deposit rates, Equation (2) imposes that no depositors are paid less than 0. This can be interpreted as the result of competition with cash, or institutional or legal restrictions.

Since I only observe deposits at the bank level, aggregated across many types of depositors paid different rates (small, large, corporate, retail, etc.), I must assume that there is some unobserved heterogeneity in rates represented by a depositor-specific error term (η). The observed deposit rate r_t^o is therefore given by:

$$r_t^o = \int_{-\infty}^{\infty} \max\left\{r_t^d + \eta, 0\right\} dF(\eta)$$

where F denotes the CDF of the distribution of the depositor rate error term η.[5]

I assume the cross-sectional distribution of deposit errors, η, is normal with variance s^2.

[5]The contemporaneous paper Xu (2024) takes a similar approach to estimating deposit rate response to interest rate changes.

Then standard properties of the normal distribution tell us that the observed rate is given by the function $g(r_t^d)$ defined by Equation (2).

Importantly, this model nests the simple constant β model used in Drechsler et al. (2023), DeMarzo et al. (2024), and other papers, if $\delta = 0$ and $s = 0$.

1.2 Estimation

I fit Equations (1) and (2) to quarterly deposit interest rate data from 1987 Q1 to 2023 Q2 using maximum likelihood estimation. Bank data are compiled from the FFIEC Call Reports.[6] For the bank-level analysis, I include data quality filters described in Appendix E.1. For quarterly risk-free rates, I use the three-month zero-coupon rate at the start of the quarter, using the Treasury curve calibrations from Gürkaynak, Sack, and Wright (2007).

I estimate the deposit rate function for interest-paying instant-access deposits only (savings accounts and interest-paying transaction accounts), covering approximately 70% of domestic deposits over 2002–2023.[7] Time deposits and non-interest-paying transaction deposits constitute another 30% of deposits and are included separately in the calculation of bank value in Section 3.

1.3 Results

Table 1 reports the estimated coefficients. The fixed spread (δ) and pass-through delay (γ) parameters are tightly identified by the data. We can reject a model with no fixed spread for interest-paying instant-access deposits with a high degree of statistical significance.

The long-term profitability and value of deposits come from a fixed spread of approximately 2%. If we assume for some back-of-the-envelope calculations that the long-term discount rate is 5%, then the value of a permanent flow of the fixed spread would be $\frac{2\%}{5\%} = 40\%$ of the total book value of interest-paying instant-access deposits. Hence the

[6] The starting point for the SAS code to download the Call Report data was taken from Alexi Savov's website. I am grateful to Alexi and coauthors for making this code available.

[7] I combine savings accounts and interest-paying transaction accounts because the regulatory distinction between these accounts disappeared during the period under study.

compression of this fixed spread as rates go to 0 will have a large impact on the value of the bank, as I will show in Section 2.

The long-term pass-through at high rates (β) does not seem to be as important. The estimate is near 1. Indeed, the model restricted to impose $\beta = 1$ shows slightly better out-of-sample prediction ability (see results in Section 1.5). I therefore use the $\beta = 1$ model as the baseline for my calculations in the remaining sections. At low rates, pass-through will be lower than 1 due to the zero floor and the cross-sectional variance.

To visualize how well this model fits the long-term response of deposit rates to risk-free rates, Figure 4 plots the relationship between observed deposit interest rates and a moving average of the risk-free rate. The two show a non-linear relationship similar to the zero-bound truncation function from the model (plotted in orange).

I also estimate bank-specific coefficients, with results summarized in Appendix E.3. The average bank results are similar to the aggregate, although with considerable cross-sectional standard deviation. High δ is correlated with high insured deposit share and high fixed costs, although, surprisingly, it is lower at banks in more concentrated local markets.

1.4 Comparison with Drechsler et al. (2021) and DeMarzo et al. (2024)

Previous work has found instead that deposits have a low β to the risk-free rate and no fixed spread. Drechsler et al. (2021) estimate that average bank interest expense has a beta of just 0.35, and DeMarzo et al. (2024) estimate that average deposit betas are approximately 0.65.[8]

My findings differ from Drechsler et al. (2021) because I measure long-term pass-through after allowing for spreads to follow an autoregressive process. Drechsler et al. (2021) instead measure short-term pass-through by regressing annual changes in deposit rates on risk-free

[8] DeMarzo et al. (2024) estimate aggregate deposit spreads of approximately $0.26 \times r^f$ per dollar of tangible assets. Deposits are about 75% of tangible assets on average during the period, implying $1 - \beta \approx 0.26/0.75 \approx 0.35$.

rates. Their average β of 0.35 resembles the coefficient of 0.4 that my model implies should be estimated by a regression of four-quarter changes in r_t^d on four-quarter changes in r_t^f. To illustrate this difference, Figure 5 plots interest-paying instant-access deposit rates against risk-free rates, with sequential points connected. The low short-term pass-through traces out a series of shallow lines with a slope of less than one-half, but the long-term trend moves up and down along a much steeper slope of roughly 1.

My estimates also differ from those of DeMarzo et al. (2024) for two main reasons. First, DeMarzo et al. (2024) study the interest expense of all deposits as a share of assets, whereas I study the interest rates on interest-paying instant-access deposits. This difference in deposit scope accounts for half the difference in estimated fixed spread. I choose my narrower scope to avoid confusing the effects of interest rates with secular trends in the deposit mix. In particular, the late 1980s and early 1990s had a large quantity of non-interest-paying deposits, most likely due to the delayed effect of the repeal of Regulation Q, increasing the measured aggregate deposit spread.

The second key difference is that my empirical model allows for non-linearity around the zero lower bound. DeMarzo et al. (2024) instead estimate a linear regression in levels for deposit rates. Because there are no observations with negative deposit rates and a number of quarters with risk-free rates near zero, a linear regression will tend to find a fixed spread close to zero.

This second difference is the key reason why our papers find different implications for the duration of the banking sector. To show this I re-estimate my non-linear model on the exact deposit scope, sample period, and specifications from DeMarzo et al. (2024) and calculate the implied effect on bank duration. The amount of non-linearity of the cash flows is lower using the all-deposit scope from DeMarzo et al. (2024), but the size of the cash flows is larger. Ultimately this specification gives similar patterns to my main results for the contribution of deposits to bank duration.

Appendix B shows a reconciliation of the sources of differences between our aggregate estimates, and calculates deposit duration under a non-linear model fitted to their sample

period and deposit scope.

1.5 Goodness of fit

The censored autoregressive model fits the data well, even far out of sample. A more standard linear model, either estimated in levels or first differences, shows much greater prediction error.

Table 2 shows the R^2 and prediction error of the fitted model for interest-paying instant-access deposit rates as well as simple linear regressions of these deposit rates on the fed funds rate in levels and differences. In sample, the R^2 of the censored autoregressive model is 0.997, making its error variance 20× lower than that of the linear regression in levels.

To ensure that I am not overfitting, I test the out-of-sample prediction with a "leave-five-year-out" and "leave-ten-year-out" approach. For each quarter, I split the data into (a) an out-of-sample dataset of the following 5 or 10 years of data and (b) an in-sample dataset of the rest. I then fit the model on the in-sample set and calculate the residuals for the last out-of-sample quarter.

The censored autoregressive model predictions hold up well even 5 years out of sample, with an R^2 of 0.95, and prediction error variance that is 3× lower than the linear model in levels and 5× lower than the linear model in differences.

2 Deposit value & duration

In this section I estimate the approximate no-arbitrage value of a permanent dollar of interest-paying instant-access deposits, applying a replicating portfolio approach to the cash flows modeled in Section 1. Conditional on the estimated deposit rate function, this approach requires no model of interest rate dynamics: the value comes directly from quoted option prices. Assumptions are only required to fill in prices for strikes or maturities of options without available quotes. I show that the value of deposit spreads declines when rates are high or low, with particularly sharp declines at low rates. I then briefly describe

the conditions under which the counterintuitive decline in spread value at low rates should arise.

2.1 Constructing an almost-replicating portfolio

Long-term rate value vs. delay value

The value of the deposit can be decomposed into the value of the "long-term rate," and the value of the "delay," where the long-term rate (r_t^l) is the deposit rate that would be observed if long-term pass-through were instantaneous (i.e., if $\gamma = 0$). I will focus in this subsection on the value of the long-term deposit rate, because the value of the delay is likely to be small, and the value of the long-term deposit rate is a function only of contemporaneous risk-free rates and so can be replicated from simple derivatives.

Defining the long-term rate as:

$$r_t^l = g(\beta(r_t^f - \delta))$$

we can write deposit cash flows and value as:

$$\underbrace{r_t^o}_{\text{Deposit rate}} = \underbrace{r_t^l}_{\text{Long-term rate}} - \underbrace{r_t^l - r_t^o}_{\text{Delay effect}}$$

$$\underbrace{PV_t(r_{t+j}^o)_{j=0}^{j=\infty}}_{\text{Deposit value}} = \underbrace{PV_t(r_{t+j}^l)_{j=0}^{j=\infty}}_{\text{Long-term rate value}} - \underbrace{PV_t\left(r_{t+j}^l - r_{t+j}^o\right)_{j=0}^{j=\infty}}_{\text{Delay value}}$$

The contribution of the delay effect to deposit spreads is positive when rates have recently risen but rapidly shrinks to zero due to the estimated pass-through delay rate of $\gamma = 0.8$ quarterly, or 0.4 annually. Since value depends on long-term cash flows, we should expect the large majority of the value to come from the long-term rate component. I confirm this in Section 3 with a model to value the whole deposit cash flow including delay effects, which shows similar size and patterns.

Deconstructing long-term rates as a sum of interest rate options

Section 1 derives the censoring function g as a sum of a range of different depositor cash flows, each of which has an option-like structure. Specifically:

$$r_t^l = g(\beta(r_t^f - \delta)) = \beta \int_{-\infty}^{\infty} \max\left\{r_t^f - (\delta - \eta/\beta), 0\right\} \frac{1}{s}\phi\left(\frac{\eta}{s}\right) d\eta$$

where ϕ is the standard normal PDF.

The term $\max\left\{r_t^f - (\delta - \eta/\beta), 0\right\}$ is the payoff of a call option on the quarterly risk-free rate r_t^f, with a strike price of $\delta - \eta/\beta$. The long-term rate cash flow can therefore be replicated by β multiplied by a sum of interest rate caps for all different strike prices, with the portfolio weights given by $\frac{1}{s}\phi(\eta/s)$, the density of depositors with the relevant fixed spread. The long-term rate value is therefore the value of a portfolio of these call options on all quarterly rates from now to the infinite future:

$$PV_t\left(r_{t+j}^l\right)_{j=0}^{j=\infty} = \beta \sum_{j=0}^{\infty} \int_{-\infty}^{\infty} C_{t,t+j}\left(\delta - \eta/\beta\right) \frac{1}{s}\phi\left(\frac{\eta}{s}\right) d\eta$$

where $C_{t,t+j}(K)$ denotes the price at time t of a call option on the quarterly interest rate expiring at time $t+j$ with strike price K.

This finding is an application of the general argument from Breeden and Litzenberger (1978) that an appropriate portfolio of put and call options can replicate any (sufficiently smooth) payoff function.

Interest rate caps

Fortunately, there is an active market for call options on quarterly risk-free rates in the form of the interest rate cap market. An interest rate cap consists of a call option on the quarterly rate for each quarter from time t until expiry. Caps represent one of the more liquid forms of fixed-income option, frequently purchased by borrowers to remove the risk of rising interest expenses. While comprehensive volume data are difficult to find, to give

an example of the size of trading volume, one report states that there were 494 trades with a total notional of $53 billion in November 2015 (Khwaja, 2015).

Daily prices gathered from quotes from interdealer brokers are readily available from Bloomberg. Specifically, Bloomberg provides cap prices with maturities from 1 to 30 years and strikes of 1%, 1.5%, 2%, 3%, 3.5%, and every 100 basis points from 4% to 14%. These quotes are available daily from 2006 onward, and with lower frequency back to 2000.[9]

Assumptions and approximations

Since cap prices are available only at a handful of strikes and out to 30-year maturities, I need to make several assumptions to calculate the price of a continuous portfolio of strikes at all maturities. First, I rule out negative interest rates. The interest rate cap with strike 0 then has cash flow r_t^f, and so a permanent zero-strike cap has market value equal to face value, 1. This assumption is conservative, as negative interest rates would increase the negative duration at low rates. The same assumption pins down the value of the depositors with a negative strike, i.e., where $\delta - \eta/\beta < 0$. Since the risk-free rate is never negative, these deposit cash flows constitute a fixed annuity (approximately 12 basis points annually), and can be valued using simple annuity formulas.

Second, for strikes between the quoted strikes, I linearly interpolate the Black implied volatilities of the quoted caps across strikes, separately for each date and maturity. Below the lowest quoted strike (1%) and above the highest (14%) I hold the implied volatility flat. This assumption ensures prices are continuous across all strikes. Practically, I price caps on a 5 basis point grid of strikes from 0 to 15%, and integrate the cap prices against the normal density of strikes using the trapezoid rule.

Third, to estimate the value beyond 30 years, I assume the forward price of each subsequent five-year segment of caps is constant. I observe the 25-year and 30-year cap prices, so I calculate the forward price of the difference between them—the 25–30 year cap seg-

[9] I drop the 10% strike, whose Bloomberg series is quoted on a different basis from the other strikes and is far out of line with the neighboring 9% and 11% quotes. Because almost no depositors have a fixed spread near 10%, this has no material effect on the results.

ment. I then assume this same forward price applies for all five-year periods after 30 years. Similarly, I assume the Treasury forward rate remains constant after 30 years, and I use this rate to discount these five-year cap segments. This amounts to an assumption that the variance of the interest-rate level has reached its steady state by 25 years into the future. If variance keeps increasing after 25 years, then the negative duration effect would become even larger, as values far into the future matter more. Appendix C shows how the results change if I instead take the less conservative assumption that Black implied volatilities stay constant after 30 years. I also include alternative figures calculated only using maturities up to 15 years, in case there are concerns about low liquidity in longer-maturity caps. These deliver greater and smaller variation in duration than the main estimates, respectively.

Finally, the quotes from Bloomberg come in the form of Black implied volatilities for LIBOR caps. I calculate the price of a Treasury-rate cap, assuming the same implied volatility applies to Treasury rates. I use the daily Treasury yield curve from Gürkaynak et al. (2007).

2.2 Results

Figure 1 plots one minus the value of the replicating portfolio (i.e., the estimated value of the spreads on a permanent dollar of deposits) against the 10-year zero-coupon Treasury yield from Gürkaynak et al. (2007).

Three observations arise from the chart. The first is that the value of deposit spreads is large. The average value of deposit spreads estimated here is 0.34 per dollar of deposits. Since interest-paying instant-access deposits are 4× book equity on average, this amounts to 1.4× total book equity. We should therefore expect any swings in its valuation to have large effects on bank wealth. This value is somewhat below the level estimated by DeMarzo et al. (2024), 1.9× book equity, so the main difference between our estimates lies in the sensitivity of this value to interest rates rather than its level.[10]

[10] DeMarzo et al. (2024) calculate value before tax is 25% of tangible assets. Adjusting for a leverage ratio of 10× and a tax rate of 25% gives deposit value of 1.9× book equity.

Second, the values show a distinct hump shape. Value is highest when the 10-year yield is between roughly 2% and 5% and declines at high or low rates. So the bank's risk and hedging incentives flip sign depending on the level of rates. When rates are very low, the pinned-rate effect described in the introduction prevails and low rates lower value. When rates are very high, the spreads are essentially fixed (i.e., $r_t^l \approx r_t^f - \delta$) and their value behaves like an ordinary fixed cash flow and declines as rates rise further.

Third, the hump shape is much steeper on the low side. So the bank's equity holders may be more concerned about interest rates in a very low rate environment as in 2021 than a very high one as in the 1990s. This pattern helps explain the common sentiment in the industry and markets that very low risk-free rates were bad for banks.

Another way to see the relationship between value and interest rates is to run rolling regressions of changes in value on changes in long-term rates. Figure 6, Panel (a) plots the coefficient from a one-year rolling regression of daily changes in the value of the deposit-replicating portfolio on daily changes in the 10-year zero-coupon yield from Gürkaynak et al. (2007), as well as the level of the 10-year zero-coupon yield. When rates are low, a rise in rates lowers the value of the deposit-replicating portfolio, whereas when rates are high the opposite occurs.

Panel (b) scales the implied effect on deposit franchise value by the ratio of interest-paying instant-access deposits to bank market capitalization. This shows how much interest rate sensitivity is added to the bank by the deposit non-linearity —i.e., the negative of deposit spreads' contribution to duration. These figures are economically large. At its peak value of approximately 40 in 2020, this implies a 1 ppt rise in rates would add 40% to bank value, purely through the deposit valuation channel.

2.3 Mechanism

A simple example, similar to the one provided in the introduction, helps illustrate the reasons behind this hump-shaped pattern. Suppose a bank has one dollar of deposits, paid rate $r^d = \max\{r^f - \delta, 0\}$. The deposit is invested into an asset paying r^f. Interest rates

are known with certainty and will remain at a level $r^f = \epsilon < \delta$ for the next T years, after which it will switch to its long-term permanent value of $r^f = 5\%$.

Because $\epsilon < \delta$, the bank is not paying any interest before time T. So any increase in the near-term risk-free rate ϵ just lowers the value of future deposit payments. Specifically, the value of a unit of permanent deposits is:

$$PV(r^d) = \left(\frac{1}{1+\epsilon}\right)^T \times \frac{5\% - \delta}{5\%}$$

This is clearly a decreasing function of ϵ, so bank value is increasing in risk-free rates.

On the other hand, the value of the assets' payment of the risk-free rate is always 1. For example, applying the standard pricing formula in this case gives:

$$PV(r^f) = \left(1 - \left(\frac{1}{1+\epsilon}\right)^T\right) \times \frac{\epsilon}{\epsilon} + \left(\frac{1}{1+\epsilon}\right)^T \times \frac{5\%}{5\%} = 1$$

So the value of the deposit spreads (and thus the bank) is $1 - PV(r^d)$ and is increasing in the risk-free rate.

However, once the near-term risk-free rate ϵ crosses above δ, the bank will start to pay deposit interest out in the near term, and any further increase will lower the value of the bank. So the value of the bank is hump-shaped with respect to the risk-free rate with a maximum at $\epsilon = \delta$. Figure 7 plots this relationship if $\delta = 2\%$ and $T = 10$.

This hump shape does not depend on the particulars of the example—it holds for a very general structure of the forward curve, when interest rates are unknown, and when there are a range of customers with different fixed spreads. It also applies to a level shift in both short- and long-term interest rates.

The two main requirements to observe this effect are that (a) the deposit rate function $f(r_t^f)$ must be convex—i.e., deposit pass-through must be increasing in the risk-free rate, and (b) the yield curve must be upward-sloping. Unfortunately, most of the existing

literature on the value and duration of the deposit franchise has assumed either a constant deposit pass-through (i.e., fixed "beta") or a constant risk-free rate and hence has not measured this hump-shaped valuation effect.

If deposit beta is constant, deposit value will be constant, since it is equivalent to a floating-rate note. If instead beta varies, but the interest rate is constant, then a constant deposit value is given by the perpetuity formula: $\frac{f(r^f)}{r^f}$. In this case a convex deposit rate will imply that deposit value is always increasing in the risk-free rate, and thus bank value decreasing as risk-free rates rise.

In Appendix D I state and prove a general statement that low enough risk-free rates will always drive up the value of a permanent call option on the risk-free rate with a positive strike price. Because any convex deposit rate function can be written as a sum of permanent call options (plus a linear component), this means low enough rates will always reduce the value of a permanent quantity of deposit spreads.

3 Bank duration and monetary policy shocks

In principle, the no-arbitrage deposit valuation effects measured in Section 2 could be drowned out by other sources of duration: the bank's assets, its fixed costs, and the delayed effect of deposit payments. In this section I therefore construct a full measure of bank duration, after accounting for the entire balance sheet. This requires more assumptions and modeling. I then validate this measure by showing that it predicts the response of bank stock prices to FOMC shocks using high-frequency identification. The bank duration measure predicts the observed effects of monetary policy with the right scale and a high degree of statistical significance.

3.1 Constructing a measure

Deposit duration

Section 2 calculates the value of the long-term deposit rate component of a permanent dollar of deposits using a no-arbitrage approach. To estimate the value of the delay component, I also posit an interest rate model, calibrated against the same cap data.

Interpreting these valuations of permanent deposits as the value of the bank and the deposit franchise also requires a few assumptions about discount rates and growth rates. In particular, I follow DeMarzo et al. (2024) in assuming that the expected growth rate of deposits equals the risk premium, so that deposit cash flows can be discounted at the risk-free rate. This means that the value of bank deposits at time t is equal to the quantity of deposits at time t multiplied by the value of a permanent dollar of deposits. The aggregate book equity of banks has grown at a 7% average rate since 1987, which is within the range of typical equity risk premium assumptions. The constant growth assumption is a reasonable approximation because deposit interest rates are simply far more variable than balances and account for the vast majority of changes in cash flows. Deposit interest rates plus a fixed growth rate account for 98% of the time-series variance in the level of interest-paying instant-access deposit cash flows since 1987.[11]

To value and calculate the duration of the deposits, I use a Brace, Gątarek, and Musiela (1997) model, also known as the "LIBOR market model," one of the most popular interest rate derivative pricing models used in banks and investment firms (Brigo and Mercurio, 2007).

The advantage of this model is that it allows me to:

1. Measure the effects of arbitrary shocks to the yield curve
2. Exactly match quoted prices of interest rate caps from 1 to 30 years at a particular strike

[11] R^2 from a regression of log dollar deposit interest expense on log deposit interest expense rate and year is 0.98.

3. Guarantee forward rates remain positive, or above some small negative number

I add two standard extensions to the model to better match quoted option prices: a "constant elasticity of variance" (CEV) and a shift of 1%.

In this model, each quarterly forward rate from 3 months to 30 years has a separate but correlated source of risk dB_t^T. Under the time-T forward measure, the forward rate expiring at time T follows the process:

$$dF_t^T = \left(F_t^T + 1\%\right)^\theta \sigma_T dB_t^T$$

The parameters of this model are:

- σ_T: volatility of the three-month forward rate expiring at time T.
- 1%: a shift parameter to allow for small negative interest rates
- θ: the CEV parameter, which governs how the instantaneous volatility of forwards relates to their level. If $\theta = 1$, the model is lognormal; if $\theta = 0$, it is normal; and if $\theta = 0.5$, it is a square-root process. This parameter is important to calculating the duration of the option—the lower the value of θ, the less the value of a call option will increase when rates rise.
- The correlation matrix of the Brownian motions for the different forward rates

Under any other measure, no-arbitrage restrictions give a drift term to each forward rate that depends on their correlation. Collectively the forward rates can vary independently, but exhibit mean reversion. More details on the functioning of the Brace et al. (1997) model can be found in fixed-income derivative textbooks, for example Brigo and Mercurio (2007) or Lesniewski (2019).

The value of call options (i.e., caplets) can be calculated in closed form. If $\theta = 1$, they are given by the Black formula. With $\theta \in (0, 1)$, they follow a slightly different expression involving the χ^2 distribution that can be found in standard textbooks. Valuing the delay

effect of deposit rates (i.e., γ from Section 1) requires a Monte Carlo simulation of interest rates.

The forward rates in each period are taken from the Gürkaynak et al. (2007) yield curve data. The correlation matrix of interest rate shocks is also calculated from the same data. As in Section 2, I assume the forward rates and forward prices of call options are constant after 30 years.[12]

I choose the values of σ in each period that best price 1–30 year interest rate caps with strikes from 1% to 3%, since this resembles the structure of the fixed spread value. Interest rate cap data are taken from the Bloomberg "Volatility Cube" and based on broker quotations.

Finally, the CEV parameter θ is calibrated from a regression of daily squared changes in forward rates on the level of forward rates. Taking the log of the squared forward rate process gives us:

$$\log\left(\left(\Delta F_t^T\right)^2\right) \approx \theta \times \log\left(\left(F_t^T + 1\%\right)^2\right) + \log\left(\sigma_T^2 \Delta t\right)$$

The parameter θ can therefore be recovered from a regression of the log of squared daily changes in forward rates on the log of the squared shifted level. The calibrated value is 0.35, implying the process is somewhere between a normal and a square-root process.

The model matches the market prices of call options whose payoffs very closely resemble deposit rates. So while it is not a perfect description of reality, I am only asking it to describe prices well locally around a specific set of quoted strikes and instruments.

[12] A short-lived dislocation in bond markets at the end of 2008 means that the estimated 30-year forward rate from Gürkaynak et al. (2007) at the end of 2008 is just 0.6%. This leads to very large estimates for perpetuity values that likely do not reflect true discount rates. I therefore exclude this single quarter from the duration charts in the rest of this section.

Total bank duration

A full picture of the effects of monetary policy on bank wealth requires us to calculate the contribution of assets and all other sources of bank income and costs.

I calculate the duration of the assets on the balance sheet using data on maturities and interest income from the Call Reports. I use standard bond pricing formulas to calculate the value and duration of non-mortgage loans and securities and use mortgage-backed security (MBS) index pricing data for mortgage loans and securities. Appendix F describes the methodology and results. In total, asset duration rises from roughly 1 year in 2002 to 2 years in 2023. Since banks are leveraged roughly 10× on average, this adds 10–20 years to bank duration.

The duration of non-asset, non-deposit cash flows is more challenging to estimate. These include costs, fees, profitability of new loans, any future governmental transfers or levies, etc. In my base case I assume that all of these sum to a perpetuity in expectation. I narrow my focus to publicly listed banks (representing roughly 80% of bank assets) and calculate the perpetuity size that delivers the market capitalization of the bank. The perpetuity duration is then my estimate of duration for the non-asset, non-deposit value. This approach of choosing a constant long-term cash flow to match equity prices is analogous to the approach used in the implied equity duration literature, specifically Dechow, Sloan, and Soliman (2004) and Weber (2018).

The estimated perpetuity payment is around -0.2% of assets pre-crisis and around -0.8% of assets post-crisis. The scale of this perpetuity is comparable to the size of non-interest income minus non-interest expense, which averaged -1% of assets post-crisis. The net impact of the change between the pre- and post-crisis levels is that estimated bank duration is lower post-crisis. This is reasonable, because the drop in market prices of banks after the financial crisis must translate to higher discount rates or lower cash flow expectations, either of which lowers duration.

I treat this perpetuity as my baseline assumption. As an alternative, I assume instead

that these remaining cash flows have zero duration (i.e., that their value does not respond to interest rates). This is closer to DeMarzo et al. (2024), who use historical regressions to estimate that the long-term value of lending spreads minus costs is around 0 for the average bank. Figure 9 compares the two. The choice of assumption leads to a substantial difference in the level of bank interest rate sensitivity, but the time-series patterns are similar. A full description of the methodology and its results is in Appendix G. I also consider an alternative specification in which I value all deposits as in DeMarzo et al. (2024), but allowing for non-linearity in the deposit rates.

For the regression analysis, I use sensitivities estimated from the last complete quarter of balance sheet data before the date of the FOMC announcement (e.g., I use March data and duration estimates for an announcement in April). Cash flow sensitivity parameters are estimated for each bank using the whole time series of data, as described in Section 1. This is meant to capture the best possible estimate of deposit sensitivity—investors may have information on deposit interest rate sensitivity available to them through annual reports, analyst calls, and customer surveys that is richer than the simple time series employed in this analysis. As a robustness test, in Appendix J I recalculate the aggregate regressions using expanding window estimates that do not use any future data.

A typical monetary shock

To predict the effect of a monetary shock on bank value I first need to know how it changes the yield curve. It is not exactly right to measure the effects of level changes to the yield curve, since monetary shocks are typically larger at the short end than at the long end of the curve (Jarocinski, 2024).

As a measure of "typical" monetary shock, I use the first principal component of shocks to four different yields, accounting for 70% of variance. More specifically, I use the fed funds futures rate and 5-, 10-, and 30-year Treasury futures, 2002 Q3–2023 Q2. Principal components of the fed funds future and the 5- and 10-year yields are often used to summarize and categorize monetary shocks (e.g., Swanson, 2021; Jarocinski, 2024). I add

the 30-year rate as well because long-term yields are particularly important for bank and deposit valuation. Shock data are taken from Gürkaynak, Karasoy-can, and Lee (2022) up to 2019 and Jarocinski (2024) thereafter. I scale the principal component such that the effect on the 5-year Treasury yield is 1.[13]

The deposit valuation described in Section 2 allows me to calculate the effect of arbitrary shocks to the yield curve. Since the high-frequency data only identify shocks to a few points on the curve, I need to fit some function to connect these points and fill in the rest of the rate shocks. I fit a Nelson–Siegel function that exactly matches the shape of the principal component.

Figure 8 shows the first principal component of observed shocks, and the fitted yield curve shocks. The shock is scaled to have size 1 at 5 years and follows the Nelson–Siegel function:

$$\text{fwd}(\tau) = \beta_0 + \beta_1 e^{-\tau/\lambda} + \beta_2 \left(\tau/\lambda\right) e^{-\tau/\lambda}$$

where τ denotes maturity in years.

I use daily bank stock returns to measure the effect of these shocks on banks. Returns data are taken from CRSP and linked to Call Report data using the RSSD ID–PERMCO linking file provided by the New York Fed. I use daily changes in bank stock prices instead of returns from the 30-minute announcement window. This adds noise but should not introduce any endogeneity problems and has the benefit of allowing for banks that might not have high liquidity during 30-minute windows.

3.2 The duration time series

Figure 9 plots the resulting estimate of the effect of monetary shocks on total bank value, under the baseline assumption that the bank's residual cash flows form a perpetuity, and under the alternative assumption that they have zero duration. The residual cash flow

[13]The long-term shocks to interest rates will be some combination of quantitative easing, forward guidance, and information effects. I do not try to separate these because my interest rate risk measures are based on no-arbitrage prices, and work equally well for risk-premium-driven shifts and expectation shifts.

assumption leads to a substantial difference in the level of bank interest rate sensitivity. But the time-series patterns are similar under the two assumptions: in both cases, the exposure of banks to a rate rise is greatest when long-term rates are at their lowest, peaking in 2020, and falls sharply as rates rise from 2022. The deposit contribution to the baseline measure, shown in Figure 17 in Appendix H, resembles the replication-based estimates from Section 2: positive when rates are low and slightly negative when rates are high.

The peak values are large relative to the value of banks—in 2020, the baseline estimates imply that a single percentage point drop in interest rates would lead to up to a 75% drop in the market value of banks (around 25% under the alternative assumption). As a comparison point, banks' market-to-book ratio fell by 60 ppt during the COVID-19 pandemic, from 2019 Q4 to 2020 Q1. The interest rate hedging incentives of banks therefore change dramatically over time.

The swings in duration are also macroeconomically large in dollar terms. At the 2020 peak, the aggregate value of banks increased by approximately $14 billion per basis point rise in rates under the baseline assumptions. By 2023, that number had dropped to $1.3 billion. For comparison, the dollar value of a basis point (DV01) of all Fed holdings as of 2023 is only roughly $4.5 billion.[14] So in a few years the swing in public bank interest rate sensitivity was more than twice as large as the duration that QE in aggregate had removed from private balance sheets.[15]

This variation in bank interest rate risk is substantially larger than other important sources of time variation in duration identified in the finance literature. Mortgage-backed security prepayment risk and insurance company convexity and duration mismatch are two key factors often credited for driving investor interest rate exposure and long-term yields (e.g., Hanson, 2014; Carboni and Ellison, 2022; Domanski, Shin, and Sushko, 2017). In Appendix I I show that swings in bank duration are larger and more persistent than either

[14] Wan and Becker (2024) estimates Fed holdings as $5.5 trillion in 10-year Treasury equivalents. I assume 10-year Treasuries have a duration of 8 for this simple calculation.

[15] Greenwald et al. (2023) make a similar comparison using their duration estimates.

of these sources, with a standard deviation roughly 3–4× higher.

3.3 High-frequency regressions

I follow the high-frequency identification literature (e.g., Nakamura and Steinsson, 2018; Gürkaynak et al., 2022) and treat 30-minute changes in interest rates around FOMC announcements as exogenous shocks to the yield curve. The identifying assumption is that, within these narrow windows, there is no other news that systematically moves bank stock prices. I then test whether the measured interest rate sensitivity of banks predicts the response of their stock prices to these shocks.

Aggregate results

Figure 2 compares the effect of interest rate shocks, measured using a rolling regression of 20 announcements (2.5 years), to my baseline measure of bank interest rate sensitivity. There is a clear and consistent relationship between my measure and the effects of shocks outside of the financial crisis period. This result is not just a peculiarity of the announcement days. In Appendix J, I repeat this plot using all days and show it follows the same pattern.

To confirm and test this relationship, Table 3 shows a time-series regression of banking sector returns on the interaction of measured interest rate risk with rate shocks. If the duration measure is accurate, then the coefficient on the interaction term should be near one and highly significant.

The results are highly significant, all with p-values below 0.01. They are also of the correct scale, with coefficients of almost exactly 1 in the specifications without controls for market returns (columns 3 and 6 add these controls, discussed below). Effects are generally more robust when the financial crisis and its immediate aftermath are excluded (columns 4–6). Bank stocks were highly volatile during this period and investors were likely to be parsing Federal Reserve announcements for information about the nature of bank assistance programs, adding noise and possibly endogeneity into the relationship.

Much of this predictive power is driven by the estimated deposit duration, although

the duration of modeled fixed costs also plays an important role. In Appendix K, I run this same prediction regression using different components of the estimated bank duration. Deposit duration alone is a significant predictor of the response of bank stocks to monetary policy shocks. However, deposits appear to deliver too little variation in duration on their own—the point estimates are (insignificantly) greater than one. Similarly, a model with only negative duration from the fixed cost perpetuity and no deposit duration predicts bank stock response to monetary policy shocks but with a coefficient significantly greater than one. In other words, banks seemed to have negative duration from both the deposit effect and fixed costs.

The effects are also quantitatively important for explaining the overall variation in the value of banks. Excluding the financial crisis, the duration and the 30-minute interest rate shock explain a full 10% of the variance in daily bank returns (see R^2 in column 4). This is not just because interest rate shocks are large on FOMC days—a simple regression of all daily stock returns on daily rate changes $\times$ duration also shows a similar R^2.

One potential concern could be that monetary shocks impact bank value through credit spreads and loan default expectations rather than interest rates. I control for this channel by adding daily changes in high-yield and investment-grade credit spreads (the ICE BofA option-adjusted spread index) as controls in columns 2 and 5, with little effect. I also control for market returns in columns 3 and 6 to demonstrate that the observed patterns reflect bank-specific effects rather than solely general market movements. Other stocks of course have interest rate risk as well, so the coefficient after controlling for market returns should be interpreted with caution—it no longer represents a pure bank interest rate risk measure.

Bank-level results

Table 4 shows the results of a regression of individual bank returns on interest rate shocks interacted with measured bank duration. The coefficient on the interaction term is again highly significant, although with a coefficient closer to 0.5 than 1 (columns 1 and 4).

To show that the effects are not purely driven by the time series, i.e., that cross-sectional differences in measured interest rate risk predict cross-sectional differences in share price response, columns 2 and 5 include time fixed effects. After excluding the volatile financial crisis period, the cross-sectional results are borderline significant, although the coefficients are well below 1, perhaps due to measurement error and the challenges of fitting a three-parameter model of deposit rates for every bank.

Just as for the time-series results, the results are not driven by credit spreads or whole market returns. Columns 3 and 6 show the results including bank-specific controls for credit spreads and market returns (i.e., interactions of bank dummies $\times$ spread changes or returns), which remain significant when excluding the financial crisis period.

To avoid the results being skewed by small banks with low daily liquidity and non-standard business models, the regressions in Table 4 are weighted by each bank's share of total sector market capitalization. I have also excluded a few banks with non-commercial-banking business models (e.g., investment banks, custodians).

Comparison with other duration measures

Other measures of bank duration in the finance literature tend to move in the opposite direction from this paper's results because they do not include the large, concave deposit value. They do not match the high-frequency shock data—in fact most tend to predict bank stock price reactions in the wrong direction.

I consider three alternative bank interest rate risk measures that I can reproduce or extract from other papers. First, I consider a duration calculation based on the franchise value calculations from DeMarzo et al. (2024). Second, I consider the measure of interest rate risk exposure of banks' fixed-income position from Begenau et al. (2015). And third, I consider the simple "income gap" measure of balance sheet duration mismatch used, for example, in Haddad and Sraer (2020). Figure 10 plots these measures together—it is clear that they reach their minima as this paper's measure reaches its maxima.

These measures move in the opposite direction from mine because they are focused on the bank's assets rather than its deposit franchise. DeMarzo et al. (2024) estimate that for the average bank in 2021, the interest-rate-sensitive portion of bank franchise value amounts to a very small perpetuity of just 0.07% of assets. So most of the interest rate exposure comes from the asset book. Begenau et al. (2015) focus on the assets and statutory liabilities in their estimates of the banks' fixed-income position. And the income gap used in Haddad and Sraer (2020) is similarly only a measure of the interest rate exposure of assets and statutory liabilities.

The time series of bank duration in this paper is negatively correlated with asset duration, and thus with these other measures, because banks accumulated more long-maturity assets over the years as interest rates declined. This mismatch is also documented in Haddad and Sraer (2020) and Begenau et al. (2015). Haddad and Sraer (2020) frame this as a puzzle for future research to resolve. Begenau et al. (2015) argue this is evidence of negative franchise duration not reflected by their statutory balance-sheet-based measure. My measure provides an explanation for the sources of this franchise duration.

I calculate the DeMarzo et al. (2024) measure by using my estimate of security duration and their estimate of the franchise perpetuity. For the Begenau et al. (2015) metric, I collect aggregate interest rate factor exposures up to 2014 from their paper (Figure 8) and divide by aggregate bank market capitalization.[16] And I calculate income gap using the standard definition applied to the aggregate bank balance sheet. The figures may not exactly match the respective authors' calculations, but they should at least be highly correlated.

In Table 5, I repeat the time-series duration regression from Table 3 for each of the alternative duration measures shown in Figure 10. In the sample excluding the financial crisis, all measures have the wrong sign—more positive measured interest rate exposure is associated with more negative response of stock prices to positive interest rate shocks—and significantly so for the DeMarzo et al. (2024) and Begenau et al. (2015) measures.

[16]Interest rate factor exposures are calculated in units of exposure to the 5-year interest rate swap. To convert to a duration measure I multiply by 5 as a rough estimate of interest rate swap duration.

This negative sign suggests that banks allow their asset duration to move in the opposite direction from their franchise duration, consistent with their time-varying hedging motive.

4 Duration and long-maturity asset holdings

Banks increased their holdings of long-term securities in 2020–21. These long-term assets were widely credited as a key cause of the regional bank crisis in 2023, and the build-up is often blamed by the press and regulators on a focus on short-term interest margins (e.g., Federal Reserve Board, 2023; Wilkes, 2023). In this section I show that these long-term security purchases can instead be interpreted as a hedge of long-term value, not just short-term interest income. They reduced stock duration and volatility when they were purchased. Furthermore, the banks exposed to the greatest negative duration from deposits were the ones that bought the most. If these assets hedged interest rate risk, then it is natural to ask why they got regional banks into such trouble. I highlight two reasons. The first is convexity—deposit value is highly non-linear in rates, so when rates rose, long-term assets ceased hedging bank value and instead added to its interest rate risk. The second is that hedging going-concern value is different from hedging liquidation value. I illustrate in this section how actions to hedge the going-concern value of the bank can increase potential losses under a bank run.

4.1 Long-term securities hedged duration

Banks bought large quantities of long-term securities during the zero rate period, and particularly when long rates were at their lowest in 2020–21. Figure 11 shows the time series of the share of bank assets invested in securities with maturities over 15 years, the longest-maturity bucket available in the Call Report data. When rates were higher, these assets (primarily MBS) constituted under 7% of assets. They rose up to 9% at the end of 2019 and then shot up to over 11% at the end of 2021. Drechsler, Savov, Schnabl, and Supera (2024) document that bank MBS constituted an increasing share of the total economy-wide mortgage holdings during this period.

This increase does not seem to be explained by credit demand, since it was driven by securities rather than lending. Nor does it seem to be consistent with the simplest versions of the "reach for yield" story (Hanson and Stein, 2015). The yield curve became flatter over this period, so there was less and less yield to reach for. From 2010 to 2020 the difference in yield between a 10-year Treasury note and a 3-month Treasury bill steadily decreased from 3.8% to −0.2%.[17]

Sections 2 and 3 show that deposit and bank duration were most negative at precisely the time when banks were accumulating these assets. This implies that far from adding to banks' interest rate exposures, these high-duration assets were actually hedging bank value by bringing their duration closer to 0.

Table 6 conducts a simpler test of this claim that does not rely on options data and estimates of deposit interest rate pass-through. I construct a value-weighted portfolio of all bank stocks and measure the volatility of its daily returns 2010–21. I then show that adding long-term Treasuries to this portfolio reduces the volatility of returns—i.e., long-maturity securities hedged the market value of banks.

Of course, after interest rates rose in 2022, the duration of deposits reversed and those long-term securities added to bank interest rate risk. Table 18 in Appendix L repeats the exercise for 2022–2025, when adding long-term Treasuries increases the volatility of the portfolio. This is consistent with the hump shape (i.e., convexity) of deposit spread value estimated in Section 2: a hedge that offsets bank interest rate risk when rates are low adds to it once rates rise.

4.2 The banks most exposed to negative deposit duration bought the most long-term securities

The cross-sectional patterns of long-term securities holdings are consistent with a story in which banks used long-term securities to reduce their negative duration from deposits. The banks with the most interest-paying instant-access deposits and with the largest fixed

[17] Data series T10Y3M from the Federal Reserve Bank of St. Louis, FRED.

spreads, whose spreads were most squeezed against 0, were precisely those that bought the most long-term securities. In Figure 12 panel (a) I show the coefficients from quarterly regressions of the share of assets held in maturities over 15 years on the estimated fixed spreads for each bank (i.e., the δ from Section 1.3). Panel (b) does the same exercise but using savings and transaction deposits as a share of assets as the regressor.

When rates were higher, there was little relationship. But as banks accumulated their long-term securities, a relationship appears, with the highest-δ and highest deposit-share banks holding the most long-term securities. At peak, in 2022 Q2, the R^2 from this single-regressor specification is 11%. To ensure that the results are not driven by small banks with little effect on aggregate securities holdings, I weight the regression by the bank's share of total banking system assets.

Drechsler et al. (2024) argue that banks' increase in MBS holdings 2019–2021 was driven by the large increase in savings and transaction deposits over the same period. Banks simply matched their new low-interest-rate-sensitivity deposit funding with low-interest-rate-sensitivity MBS.

This explanation is not inconsistent with this paper—if banks are hedging their interest rate risk, then they should care about both the changes in deposit quantity and the changes in deposits' interest-rate-risk characteristics. The interest rate sensitivity of each dollar of deposits changes in a direction that strengthens the case for long-term assets when rates are low, whether banks are hedging income or value.

This dual explanation seems to fit with the available evidence. The increase in the holdings of the longest-maturity securities was even larger than the growth in savings and transaction accounts. From 2015 Q4 to 2021 Q2 the ratio of securities with maturities over 15 years to savings and transaction deposits rose from 12.5% to 16.5%.

Additionally, the relationship between deposit quantities and long-term security holdings strengthened as rates fell. This change is easily explained if the interest-rate sensitivity of those deposits changed, as described in this paper—i.e., the hedging motive per dollar of

deposits strengthened. Figure 12 panel (b) plots the coefficient from quarterly regressions of holdings of securities with maturities over 15 years on savings and transaction deposits. Before 2010 there is a weak and usually insignificant relationship. After 2010, the relation becomes stronger, peaking in 2021.

4.3 Hedging going-concern value can increase the volatility of liquidation value

If long-term securities hedged bank value, why did they leave regional banks so exposed in 2023? The duration estimates of Section 2 give one reason: because deposit value is hump-shaped in rates, the hedge reversed once rates rose. A second reason is that the duration measured in this paper is the interest rate sensitivity of the bank's going-concern value, which includes the deposit franchise. The franchise is lost if depositors run (Drechsler et al., 2023; Haddad et al., 2023). In a run, what matters instead is liquidation value: the value of the bank's assets net of the face value of its deposits. Any asset duration added to hedge going-concern value also raises the sensitivity of liquidation value to rate rises. Drechsler et al. (2023) and Haddad et al. (2023) describe this trade-off. The estimates in this paper show that it was large at the rates of 2020–21.

A stylized bank calibrated to the 2021 estimates shows the size of the trade-off. The bank has \$1 of deposits, \$1.1 of assets and \$0.1 of book equity. Asset duration is 2, in line with Section 3, so a 1 ppt rise in rates lowers asset value by 0.022. Near its 2021 trough, the coefficient in Figure 6 panel (a) implies that the same rise raises the value of deposit spreads by roughly 0.06 per dollar of deposits. The bank therefore has negative duration: a 1 ppt rise in rates raises its going-concern value by $0.06 - 0.022 = 0.038$. To neutralize this, the bank must raise asset duration to $0.06/(1.1 \times 0.01) \approx 5.5$, which requires a large purchase of long-term securities.

Table 7 compares the bank before and after this hedge. Hedging removes the bank's going-concern exposure, but nearly triples the loss in liquidation value from a 1 ppt rise in rates, from 0.022 to 0.06. The rise in rates that wipes out the bank's book equity in

liquidation falls from about 4.5 ppt to about 1.7 ppt. The 10-year yield rose by roughly 2 ppt between the end of 2021 and early 2023. A rise of that size leaves the unhedged bank with positive liquidation value, but leaves the hedged bank unable to repay its depositors in full if they withdraw. In that case the bank is solvent only if its depositors stay, and a self-fulfilling run becomes possible (Diamond and Dybvig, 1983).

The same trade-off is large for the banking system as a whole. At the 2020 peak, the aggregate value of public banks rose by approximately $14 billion for each basis point rise in rates (Section 3.2). A 10-year Treasury loses roughly $0.8 million per $1 billion of face value for each basis point rise in rates, so neutralizing this exposure would have required roughly $17 trillion of additional 10-year Treasury equivalents (or around $6 trillion under the alternative assumption that the bank's residual cash flows have zero duration). The long-term securities that banks actually bought in 2020–21 therefore hedged only a fraction of their going-concern exposure. Even so, they were enough to produce the losses in liquidation value that left banks exposed in 2023.

5 Conclusion

Are very low interest rates bad for bank value? Under the standard fixed "deposit beta" model, it is hard to see why they would be. Deposit spreads then behave like a floating-rate note, so their value does not depend on rates, and lower rates only raise the value of the bank's fixed-rate assets (DeMarzo et al., 2024). This paper shows that the answer can be yes. Once pass-through is allowed to vary with the level of rates, as it does in the data, deposit spreads alone can lose value as rates fall, even before considering fixed costs or deposit runoff.

I find that long-term deposit spreads are approximately fixed when rates are high, and are squeezed toward zero when rates are low. Valuing these spreads with a portfolio of interest rate caps shows that deposit franchise value is hump-shaped in the level of rates, and falls steeply as long-term rates approach zero. When rates are very low, the deposit franchise therefore over-hedges the bank's assets, and banks gain from a rise in rates. At

the 2020 peak, my baseline estimates imply that a 1 ppt fall in rates would have lowered the market value of banks by up to 75%. This measure predicts the response of bank stock prices to FOMC shocks with a coefficient close to one, while existing balance-sheet measures predict it with the wrong sign. The level of measured duration depends on assumptions about the bank's residual cash flows, but its time-series pattern does not.

This time variation gives banks a strong motive to add duration when rates are low, and banks did so. Holdings of long-term securities rose sharply in 2020–21, most of all at the banks with the largest fixed spreads and the most interest-paying instant-access deposits, and these holdings reduced the volatility of bank stock returns. The purchases therefore look less like a myopic focus on short-term income than a partial hedge of bank value.

These findings also suggest that duration should not be treated as the "correct" measure of bank interest rate risk, with income-based measures and stress tests as cruder approximations to it. Duration is a local measure; it is highly sensitive to assumptions about long-term cash flows, and it values a deposit franchise that is lost if depositors run. A bank hedging its value at the low rates of 2020–21 would have bought more long-term securities, not fewer. Those securities added to its interest rate risk once rates rose, and they made it more vulnerable to a run. Managing bank interest rate risk therefore requires more than hedging the sensitivity of the bank's stock price.

References

BCBS, 2016, Interest rate risk in the banking book .

Begenau, Juliane, Monika Piazzesi, and Martin Schneider, 2015, Banks' risk exposures, *NBER Working Paper* 21334.

Begenau, Juliane, and Erik Stafford, 2018, Do banks have an edge?, *Harvard Business School Working Paper* 18.

Brace, Alan, Dariusz Gątarek, and Marek Musiela, 1997, The market model of interest rate dynamics, *Mathematical Finance* 7, 127–155.

Breeden, Douglas T., and Robert H. Litzenberger, 1978, Prices of state-contingent claims implicit in option prices, *The Journal of Business* 51, 621–651.

Brigo, Damiano, and Fabio Mercurio, 2007, *Interest Rate Models - Theory and Practice* (Springer Finance).

Carboni, Giacomo, and Martin Ellison, 2022, Preferred habitat and monetary policy through the looking-glass, *ECB Working Paper* 2022.

Dechow, Patricia M., Richard G. Sloan, and Mark T. Soliman, 2004, Implied equity duration: A new measure of equity risk, *Review of Accounting Studies* 9, 197–228.

DeMarzo, Peter, Arvind Krishnamurthy, and Stefan Nagel, 2024, Interest rate risk in banking, *Working paper* .

Di Tella, Sebastian, and Pablo Kurlat, 2021, Why are banks exposed to monetary policy?, *American Economic Journal: Macroeconomics* 13, 295–340.

Diamond, Douglas, and Philip Dybvig, 1983, Bank runs, deposit insurance, and liquidity, *Journal of Political Economy* .

Dick, Astrid A., 2008, Demand estimation and consumer welfare in the banking industry, *Journal of Banking & Finance* 32, 1661–1676.

Doherty, Katherine, 2023, Bank of america's wrong-way rate bet hurts moynihan's growth pledge, *Bloomberg* .

Domanski, Dietrich, Hyun Song Shin, and Vladyslav Sushko, 2017, The hunt for duration: Not waving but drowning?, *IMF Economic Review* 65, 113–153.

Drechsler, Itamar, Alexi Savov, and Philipp Schnabl, 2021, Banking on deposits: Maturity transformation without interest rate risk, *The Journal of Finance* 76, 1091–1143.

Drechsler, Itamar, Alexi Savov, Philipp Schnabl, and Dominik Supera, 2024, Monetary policy and the mortgage market, *working paper* .

Drechsler, Itamar, Alexi Savov, Philipp Schnabl, and Olivier Wang, 2023, Banking on uninsured deposits, Working Paper 31138, National Bureau of Economic Research.

English, William B., Skander J. Van den Heuvel, and Egon Zakrajček, 2018, Interest rate risk and bank equity valuations, *Journal of Monetary Economics* 98, 80–97.

Federal Reserve Board, 2023, Review of the federal reserve's supervision and regulation of silicon valley bank.

Gormsen, Niels Joachim, and Eben Lazarus, 2026, Interest rates and equity valuations, Working Paper 34814, National Bureau of Economic Research.

Greenwald, Emily, Sam Schulhofer-Wohl, and Joshua Younger, 2023, Deposit convexity, monetary policy, and financial stability, *Federal Reserve Bank of Dallas Working Paper* 2315.

Gürkaynak, Refet, Hatice Gökçe Karasoy-can, and Sang Seok Lee, 2022, Stock market's assessment of monetary policy transmission: The cash flow effect, *The Journal of Finance* 77, 2375–2421.

Gürkaynak, Refet S., Brian Sack, and Jonathan H. Wright, 2007, The u.s. treasury yield curve: 1961 to the present, *Journal of Monetary Economics* 54, 2291–2304.

Haddad, Valentin, Barney Hartman-Glaser, and Tyler Muir, 2023, Bank fragility when depositors are the asset, *Working Paper* .

Haddad, Valentin, and David Sraer, 2020, The banking view of bond risk premia, *The Journal of Finance* 75, 2465–2502.

Hannan, Timothy H, and Allen N Berger, 1991, The Rigidity of Prices: Evidence from the Banking Industry, *American Economic Review* 81, 938–945.

Hanson, Samuel G., 2014, Mortgage convexity, *Journal of Financial Economics* 113, 270–299.

Hanson, Samuel G., and Jeremy C. Stein, 2015, Monetary policy and long-term real rates, *Journal of Financial Economics* 115, 429–448.

He, Zhiguo, Arvind Krishnamurthy, and Konstantin Milbradt, 2026, Interest rate risk and bank hedging, *Working Paper* .

Huber, Max, 2022, Regulation-induced interest rate risk exposure, *Working Paper* .

Jarocinski, Marek, 2024, Estimating the fed's unconventional policy shocks, *Journal of Monetary Economics* 144, 103548.

Kang-Landsberg, Alena, Stephan Luck, and Matthew Plosser, 2023, Deposit betas: Up, up, and away?, *Liberty Street Economics, Federal Reserve Bank of New York* .

Khwaja, Amir, 2015, Cap and floor option volumes, *Clarus FT* .

Koont, Naz, Tano Santos, and Luigi Zingales, 2025, Destabilizing digital bank walks, *Working paper* .

Lesniewski, Andrew, 2019, Interest rate and credit models.

Lu, Xu, and Lingxuan Wu, 2025, Banking on inattention, *Working paper* .

McFadden, Daniel, 1973, Conditional logit analysis of qualitative choice behavior, *Frontiers of Econometrics* 105–142.

Moench, Emanuel, James Vickery, and Diego Aragon, 2010, Why is the market share of adjustable-rate mortgages so low?, *Federal Reserve Bank of New York: Current Issues in Economics and Financ* 16.

Morgan Stanley & Oliver Wyman, 2022, Climate, crypto, and competing in this cycle, *Corporate & Investment Banking Blue Paper* .

Nakamura, Emi, and Jon Steinsson, 2018, High-Frequency Identification of Monetary Non-Neutrality: The Information Effect*, *The Quarterly Journal of Economics* 133, 1283–1330.

Nichols, Chris, 2023, Fair value accounting and silicon valley bank failure, *South State Correspondent Division Banker to Banker* .

Peters, Sandy, 2023, The svb collapse: Fasb should eliminate "hide-'til-maturity" accounting, *CFA Institute Market Integrity Insights* .

Swanson, Eric T., 2021, Measuring the effects of federal reserve forward guidance and asset purchases on financial markets, *Journal of Monetary Economics* 118, 32–53.

Wan, Simon, and Tom Becker, 2024, Qt-lite: Quantitative tightening's limited impact, *Blackrock Insights* .

Wang, Olivier, 2020, Banks, low interest rates, and monetary policy transmission, Working Paper Series 2492, European Central Bank.

Wang, Yifei, Toni M. Whited, Yufeng Wu, and Kairong Xiao, 2022, Bank market power and monetary policy transmission: Evidence from a structural estimation, *The Journal of Finance* 77, 2093–2141.

Weber, Michael, 2018, Cash flow duration and the term structure of equity returns, *Journal of Financial Economics* 128, 486–503.

Wilkes, Samuel, 2023, Missing basel metric could have revealed svb risks, *Risk.net* .

Xu, Deheng, 2024, Interest Rates, Banks' Market Power, and Their Asset Maturity, *Working Paper* .

Figures

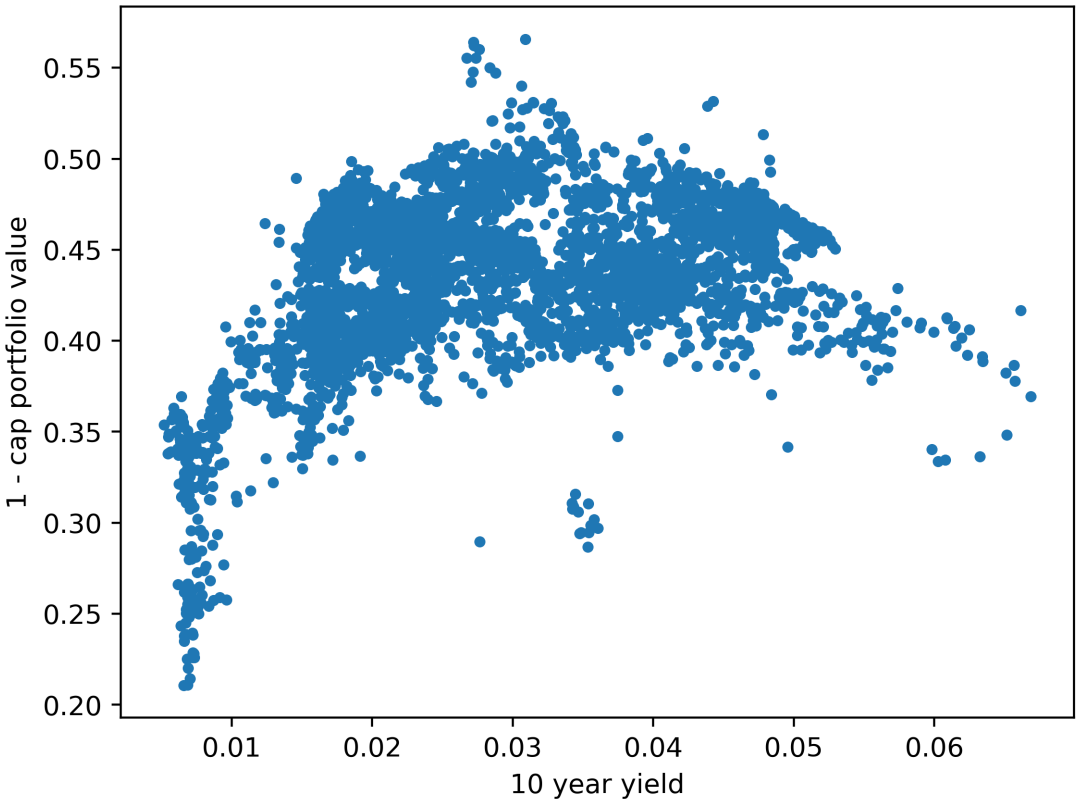


Figure 1: **Value of the deposit-replicating portfolio as a function of the interest rate**. Each point plots, for a single date, one minus the estimated value of a permanent portfolio of interest rate caps per dollar of interest-paying instant-access deposits, against the 10-year zero-coupon Treasury yield from Gürkaynak et al. (2007). The interest rate cap portfolio includes options with different strikes chosen to replicate deposit interest rates as a function of the risk-free rate. Details of the estimation are provided in Section 2.

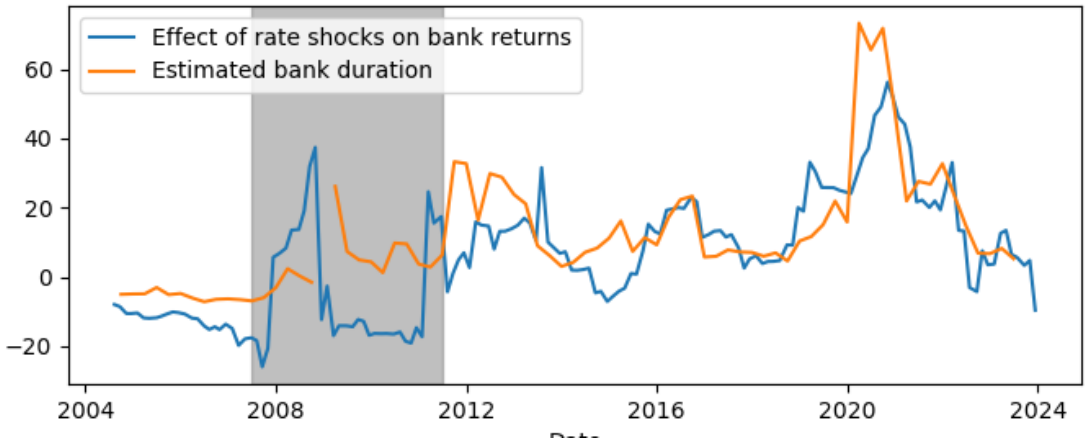


Figure 2: **Predicted (orange) vs. observed (blue) effect of rate shocks on bank values.** The orange line shows the predicted interest rate sensitivity of banks (i.e., negative duration) under the baseline assumptions described in Section 3 and Appendix G. The blue line shows the effects of high-frequency monetary shocks on bank returns, measured by a rolling regression over windows of 20 FOMC announcements (roughly 2.5 years; details in Section 3). The financial crisis (2007 H2–2009 H1) is shaded in gray.

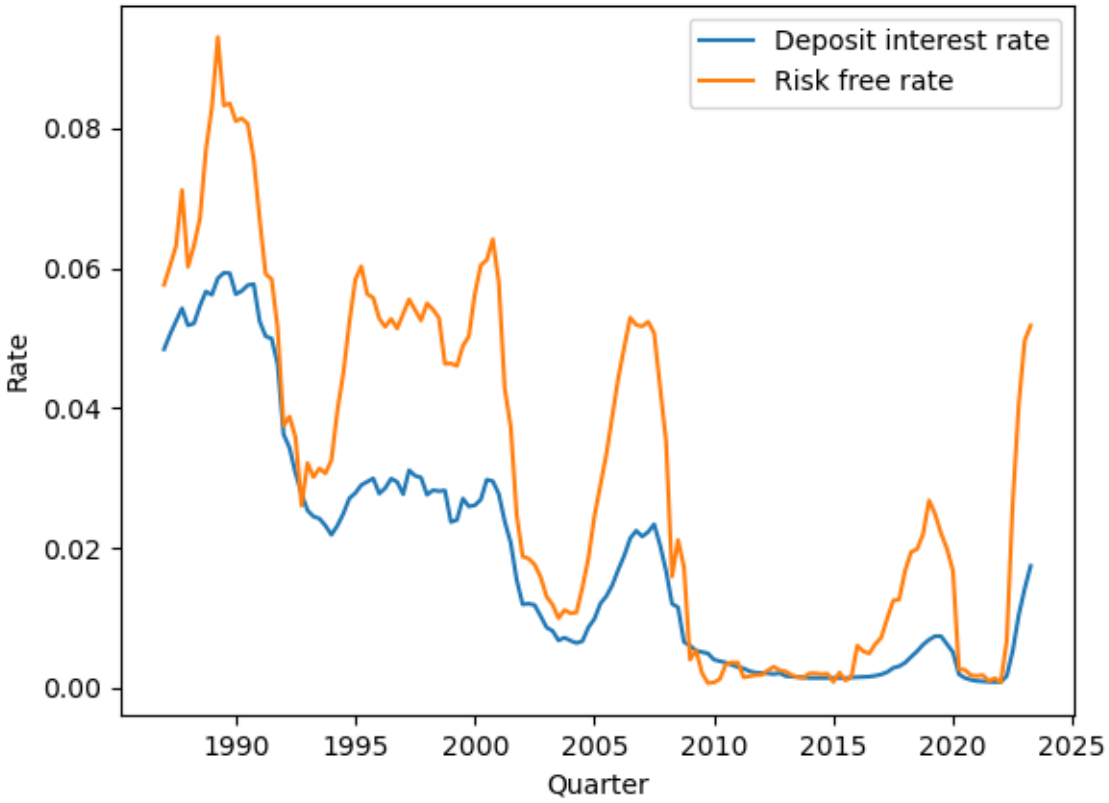


Figure 3: **Deposit rate and risk-free rate over time.** Aggregate quarterly rate of interest income on interest-paying savings and transaction deposits from the FFIEC Call Reports vs. three-month risk-free spot rate from Gürkaynak et al. (2007), 1987–2023.

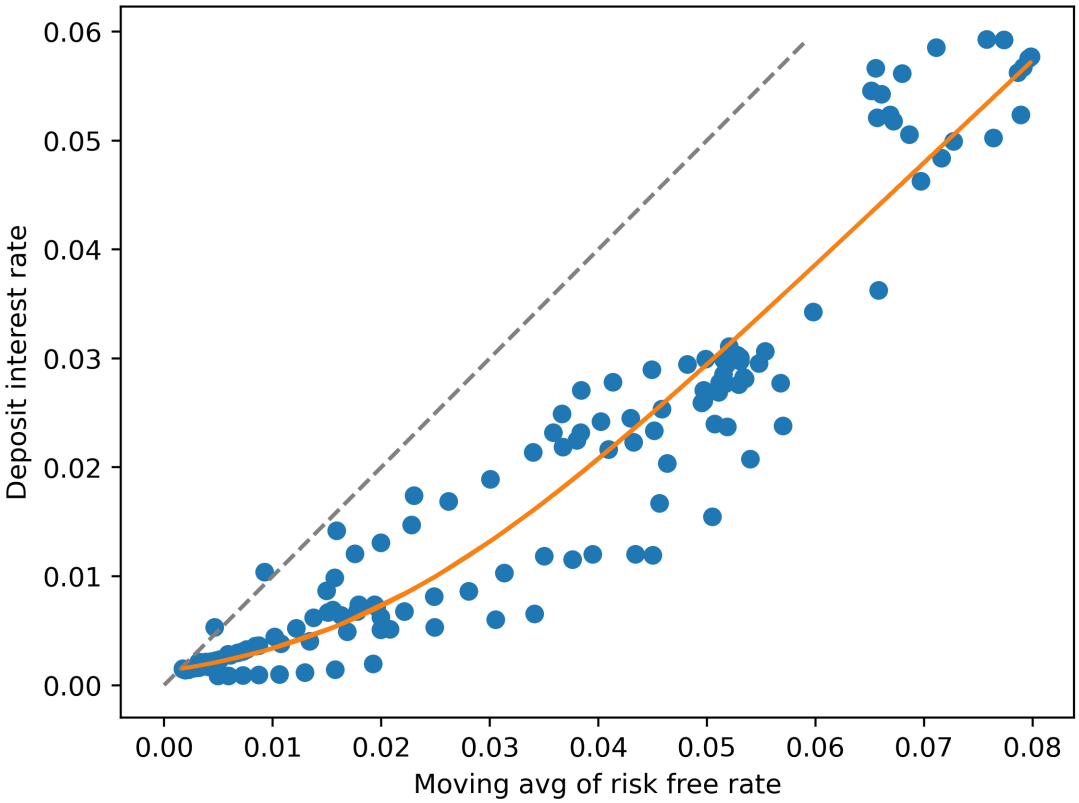


Figure 4: **Long-term relationship of deposit rates and risk-free rates.** The blue dots plot the relationship between a moving average of risk-free rates, exponentially weighted at $\hat{\gamma} = 0.788$, and observed deposit rates. The orange line plots the estimated relationship that should be observed if error terms are 0. The model implies that deposit rates are a non-linear function of the moving average of risk-free rates plus the accumulated error terms, $r_t^o = g\left(\beta\left(\overline{r^f}_t - \delta\right) + \overline{\varepsilon}_t\right)$, where $\overline{r^f}_t = (1-\gamma)\sum_{j=0}^{\infty}\gamma^j r_{t-j}^f$ and $\overline{\varepsilon}_t = \sum_{j=0}^{\infty}\gamma^j \varepsilon_{t-j}$.

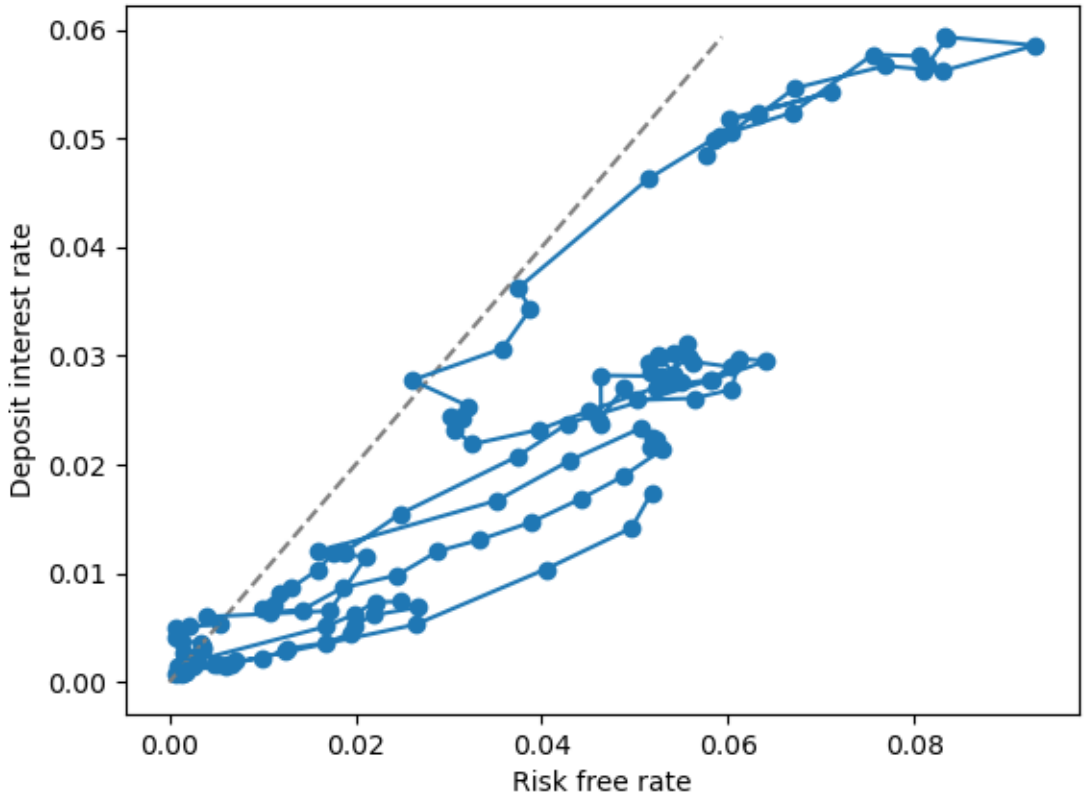
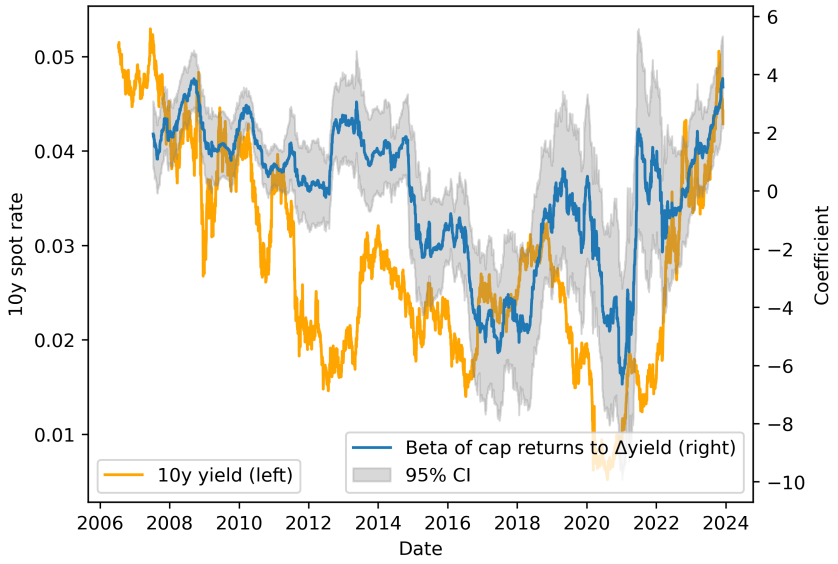


Figure 5: **Deposit rates plotted against the risk-free rate, connected in time order.** Aggregate quarterly rate of interest expense on interest-paying instant-access deposits from the FFIEC Call Reports vs. quarterly risk-free rates, 1987–2023. The dotted line shows the 45-degree line where $r^o = r^f$.

Panel (a): Rolling beta of deposit-replicating portfolio to risk-free rate changes



Panel (b): Implied deposit contribution to bank interest rate sensitivity (i.e., negative of duration)

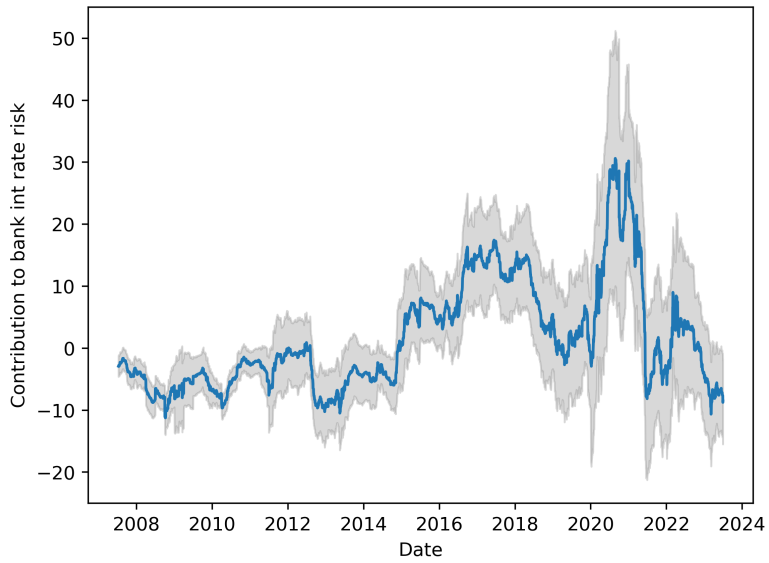


Figure 6: **Deposit effects on bank duration.** Panel (a) plots the coefficients from a rolling one-year regression of daily changes in the value of a portfolio of caps designed to replicate the cash flows of a permanent dollar of interest-paying instant-access deposits on daily changes in the 10-year zero-coupon Treasury yield. The level of the 10-year yield is also plotted. Panel (b) flips the sign of this coefficient and scales it by the ratio of aggregate interest-paying instant-access deposits to market capitalization. This represents the implied effect of a rate rise on the value of banks through the deposit valuation channel, i.e., the negative of the contribution of deposit franchise value to bank duration. Coefficients are estimated from 2007–2023.

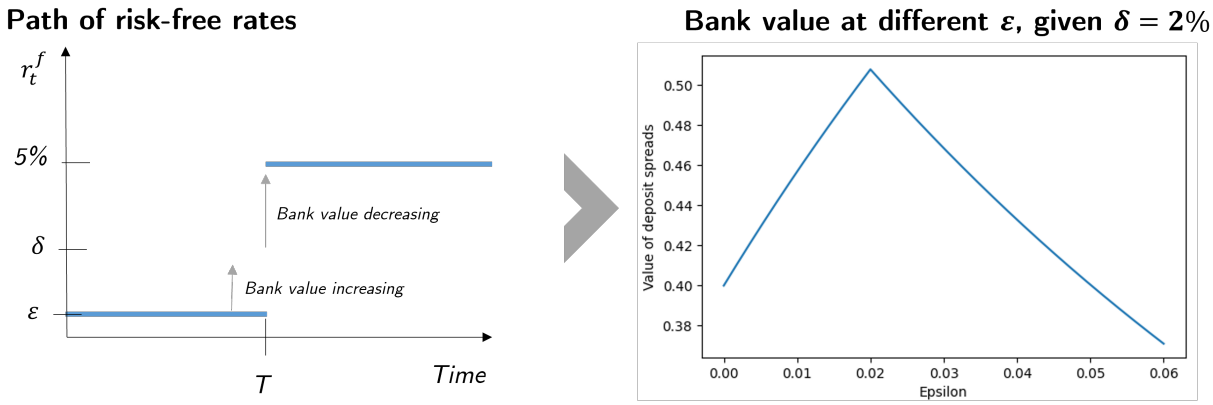


Figure 7: **Illustrative example of bank value under changes in the risk-free rate.** The left-hand panel shows the path of interest rates in the example described in Section 2.3. The right-hand panel shows the value of the bank if $\delta = 2\%$ and $T = 10$.

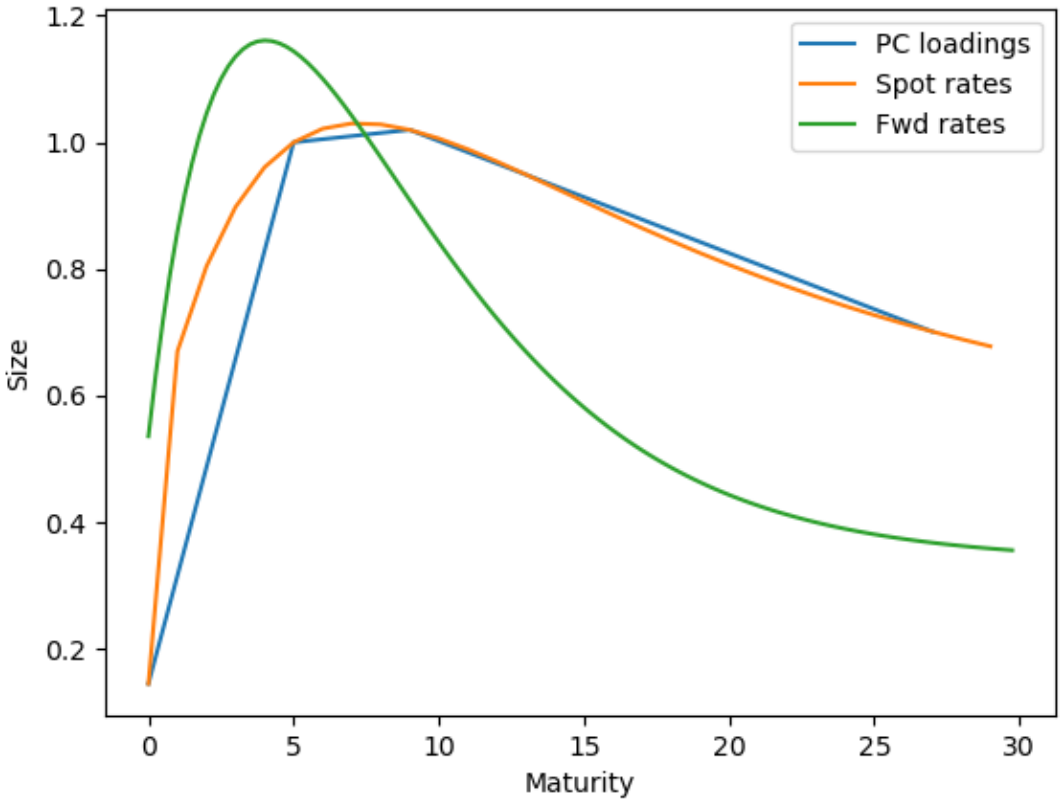


Figure 8: **First principal component of rate shocks, and fitted shock to forward rates**. The blue line plots the first principal component of high-frequency shocks to the fed funds futures rate and the 5-, 10-, and 30-year Treasury futures rates. The orange line plots the fitted shock to the spot curve and the green line plots the fitted shock to the forward curve.

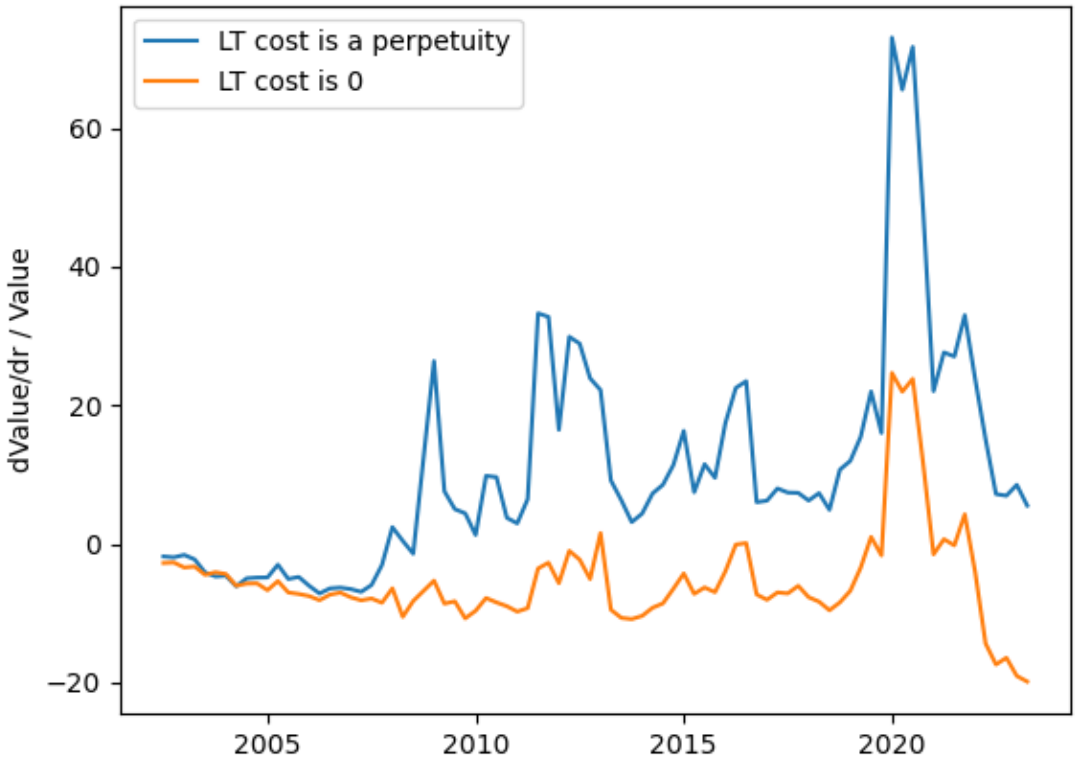


Figure 9: **Aggregate interest rate sensitivity (negative duration) of public banks under alternative assumptions for the residual cash flows, 2002 Q3–2023 Q2**. This figure shows the derivative of bank value with respect to a monetary shock, divided by market capitalization (i.e., negative duration), of all publicly listed US banks. The blue line shows the baseline, in which the residual cash flows (costs, fees, profits on new lending, etc.) are a perpetuity sized to match market capitalization. The orange line assumes instead that the residual cash flows have zero duration. Details are in Appendices G and H.

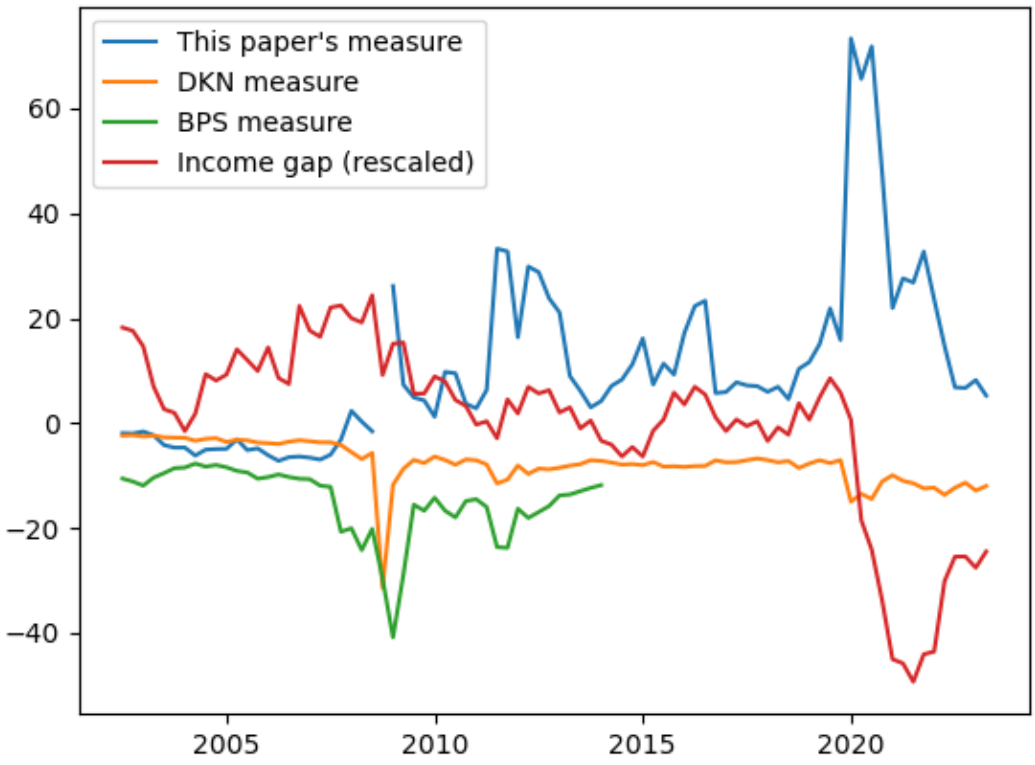


Figure 10: **Different bank interest rate risk measures**. This chart shows calculated interest rate sensitivity (i.e., negative duration) estimated from this paper (baseline; see Appendix G), and measures based on DeMarzo et al. (2024), Begenau et al. (2015), and Haddad and Sraer (2020). The income gap measure is rescaled to match the mean and variance of this paper's measure.

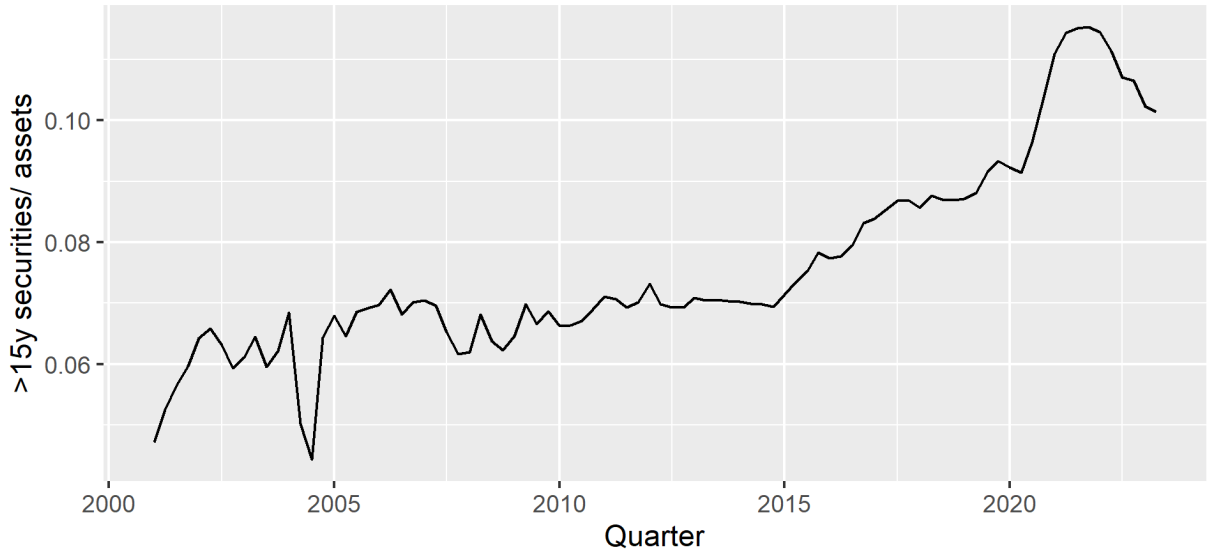
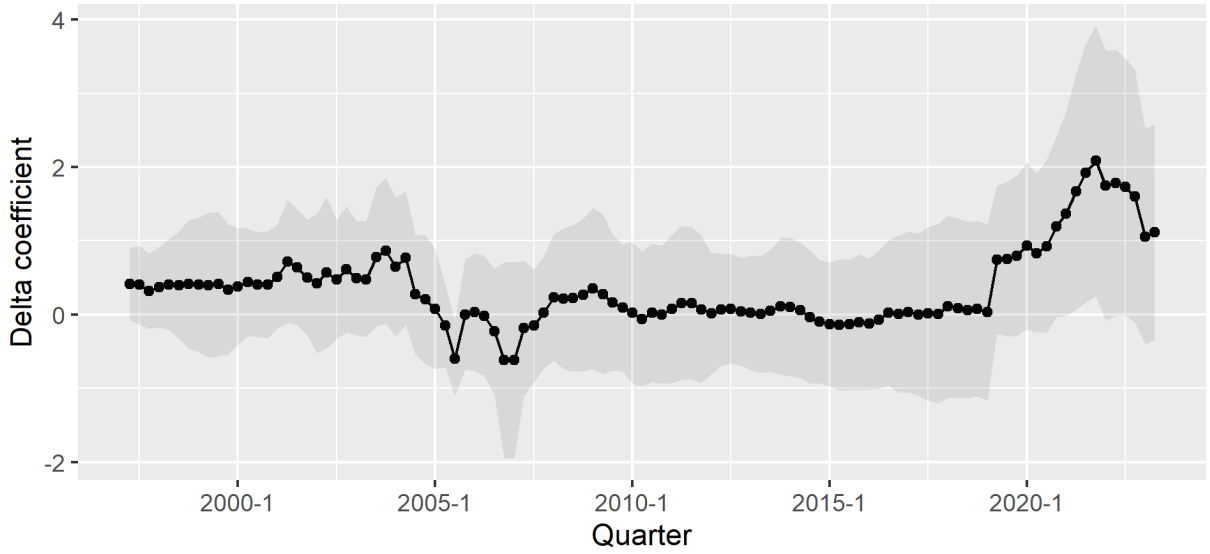


Figure 11: **Total securities with maturities over 15 years as a share of all bank assets, 1997–2023**. All commercial banks from the FFIEC Call Reports.

Panel (a): Coefficient on estimated interest-paying instant-access deposit δ



Panel (b): Coefficient on demand deposits/ assets

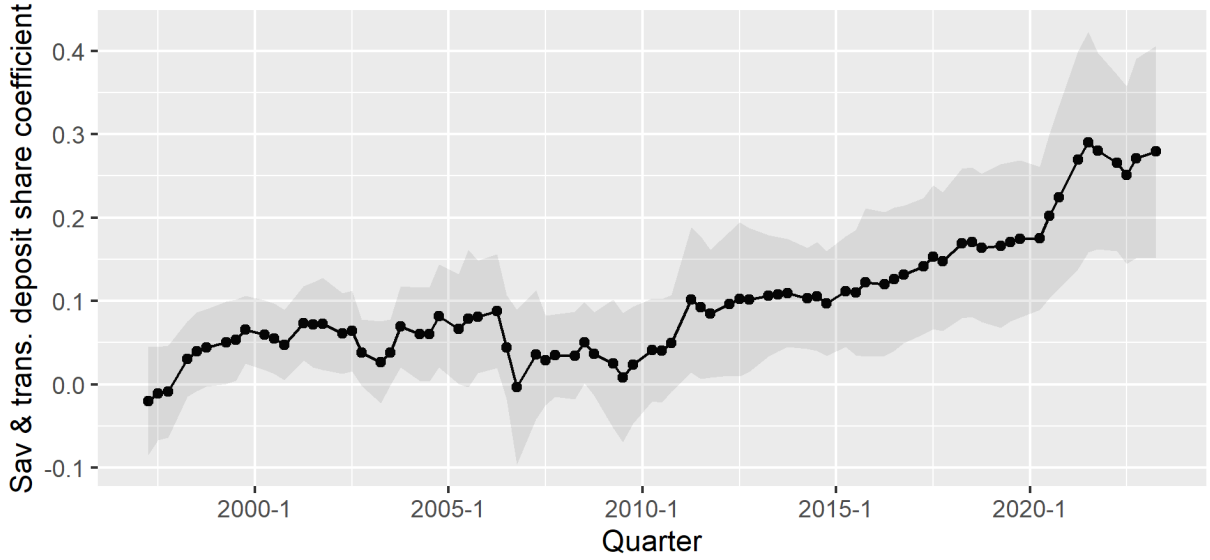


Figure 12: **Coefficients from quarterly regressions of securities with maturities over 15 years as a share of assets on explanatory variables, all banks 1997–2023.** Regressions are weighted by share of total assets, and standard errors are heteroskedasticity-consistent. Estimated δ is winsorized at the 1% level.

Tables

Coefficient & description	Beta $\neq 1$	Beta $= 1$
δ Fixed spread	0.019	0.022
	(0.005)	(0.005)
β Long-term pass-through	0.940	1
	(0.110)	
γ Pass-through delay	0.788	0.788
	(0.031)	(0.040)
s Zero bound phase-in rate	0.017	0.019
	(0.004)	(0.006)
σ Error standard deviation	0.002	0.002
	(0.000)	(0.000)

Table 1: **Aggregate deposit parameter estimates.** Estimates from maximum likelihood estimation of model specified in Section 1.1 on the time-series of quarterly aggregate commercial-bank data from US Call Reports, 1987–2023. The right column shows the estimation restricted to impose a beta of 1. Numbers in parentheses show GMM Newey–West standard errors with 4 lags.

(a) In sample

	R^2	MAPE (ppt)
Censored AR model	0.997	0.102
Linear model	0.932	0.512

(b) Five-year out-of-sample

	R^2	MAPE (ppt)
Censored AR model	0.951	0.268
Censored AR model ($\beta \neq 1$)	0.950	0.275
Linear model (levels)	0.852	0.486
Linear model (differences)	0.726	0.597

(c) Ten-year out-of-sample

	R^2	MAPE (ppt)
Censored AR model	0.938	0.248
Censored AR model ($\beta \neq 1$)	0.936	0.252
Linear model (levels)	0.807	0.430
Linear model (differences)	0.304	0.947

Table 2: **In-sample and out-of-sample goodness-of-fit statistics for linear models and the censored autoregressive model from Section 1.1**. Panel (a) provides the simple in-sample R^2 and mean absolute error (i.e., the average of the absolute value of fitted deposit rates minus observed deposit rates). Panel (b) provides results from a leave-five-year-out analysis, using the residuals from the last quarter of the five-year out-of-sample period. Panel (c) provides the same for 10 years. The first 5 years of data are dropped from panel (b), and the first 10 from panel (c).

	Full Sample			Fin. Cris. excluded		
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Rate sens. × rate shock	1.274**	1.082**	0.442**	1.169***	1.194***	0.607***
	(0.416)	(0.330)	(0.152)	(0.313)	(0.350)	(0.163)
Rate shock	−7.417	−9.214+	2.934	−4.650+	−6.172*	2.404
	(7.432)	(5.239)	(2.375)	(2.377)	(2.771)	(2.077)
Rate sensitivity	0.003	0.005	−0.002	0.005	0.004	−0.001
	(0.008)	(0.007)	(0.006)	(0.007)	(0.007)	(0.006)
ΔIG spread		23.603***	12.062***		11.783	10.133
		(4.849)	(2.533)		(16.912)	(9.300)
ΔHY spread		−10.219*	−2.889		−3.348	−2.383
		(5.148)	(1.869)		(3.786)	(1.738)
Market return			1.527***			1.249***
			(0.144)			(0.117)
Intercept	0.179	0.013	−0.132	0.103	0.095	−0.057
	(0.178)	(0.202)	(0.081)	(0.128)	(0.126)	(0.071)
Num.Obs.	168	168	168	152	152	152
R2	0.062	0.197	0.839	0.104	0.117	0.735

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 3: **Aggregate measured duration predicts bank stock price response to interest rate shocks.** This table shows results from regressions of banking sector returns on FOMC announcement days on monetary shocks to long-term interest rates, 2002 Q3–2023 Q2. "Rate sensitivity" is the measured interest rate exposure of the aggregate banking system under the baseline assumptions, described in Section 3 and Appendix G. "Rate Shock" is the first principal component of 30-minute changes in yields around FOMC announcements, scaled to a unit change in the 5-year Treasury yield (Section 3), using data from Gürkaynak et al. (2022) and Jarocinski (2024). "Δ IG spread & HY spread" denote controls for daily changes in the ICE BofA option-adjusted investment-grade or high-yield credit spread index. "Market return" is a control for the CRSP total market daily return. All columns exclude duration calculated using 2008 Q4 and columns 4–6 exclude 2007 H2–2009 H1. All standard errors are Newey–West with 3 lags.

	Full Sample			Fin. Cris. excluded		
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Rate sens. × rate shock	0.523**	0.071+	0.067+	0.684**	0.125+	0.119*
	(0.180)	(0.038)	(0.035)	(0.207)	(0.068)	(0.059)
Rate shock	−5.271			−3.837		
	(6.645)			(2.873)		
Rate sensitivity	0.003	0.003	0.004	0.005	0.003	0.005
	(0.009)	(0.003)	(0.003)	(0.010)	(0.003)	(0.004)
Intercept	0.302+			0.117		
	(0.174)			(0.134)		
Num.Obs.	55 230	55 230	55 230	49 698	49 698	49 698
R2	0.027	0.768	0.804	0.047	0.736	0.765
R2 Within		0.002	0.003		0.003	0.004
FE: Date		X	X		X	X
FE: Bank			X			X
Bank × spread & market			X			X

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Table 4: **Value-weighted duration of all public banks predicts stock price response to interest rate shocks.** This table shows results from value-weighted regressions of the returns of all commercial banks on FOMC announcement days on monetary shocks to long-term interest rates, 2002 Q3–2023 Q2, weighted by share of total market capitalization. Banks with non-commercial business models are excluded as described in Section 3.3 and the rate sensitivity measure is winsorized at 1%. "Rate sensitivity" is the measured interest rate exposure of each bank under the baseline assumptions, described in Section 3 and Appendix G. "Rate Shock" is the first principal component of 30-minute changes in yields around FOMC announcements, scaled to a unit change in the 5-year Treasury yield (Section 3), using data from Gürkaynak et al. (2022) and Jarocinski (2024). Date fixed effects are included in columns 2, 3, 5, and 6. "Bank × spread & market" denotes that controls have been added for bank id interacted with daily changes in IG and HY credit spreads (using the ICE BofA index) and CRSP total market returns. All columns exclude duration data calculated using 2008 Q4 and columns 4–6 exclude 2007 H2–2009 H1. All standard errors are clustered by date.

Panel (a): Full sample

	DKN measure	DKN (eq wtd)	BPS measure	Income gap
Rate sens. × rate shock	0.330	0.098	0.443	−3.826
	(0.456)	(0.158)	(0.311)	(3.670)
Rate shock	−1.117	−1.402	−2.347	119.057
	(9.615)	(3.700)	(6.992)	(117.178)
Rate sensitivity	−0.184	−0.065	−0.146*	0.029
	(0.115)	(0.041)	(0.060)	(0.096)
Intercept	−1.009	−0.260	−1.495+	−0.609
	(0.821)	(0.291)	(0.763)	(3.060)
Num.Obs.	170	152	96	170
R2	0.107	0.055	0.128	0.032

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Panel (b): Financial crisis period excluded

	DKN measure	DKN (eq wtd)	BPS measure	Income gap
Rate sens. × rate shock	−3.260**	−1.705*	−2.654***	−0.368
	(1.031)	(0.786)	(0.579)	(1.876)
Rate shock	−18.066***	−9.917*	−35.775***	13.368
	(4.964)	(4.014)	(5.917)	(60.853)
Rate sensitivity	0.000	−0.021	−0.044	−0.045
	(0.044)	(0.043)	(0.030)	(0.082)
Intercept	0.165	−0.076	−0.186	1.637
	(0.281)	(0.283)	(0.358)	(2.678)
Num.Obs.	152	137	78	152
R2	0.059	0.031	0.156	0.005

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Table 5: **Alternative interest rate risk measures do not predict bank stock price reaction to interest rate shocks with the right sign.** This table repeats the regressions in Table 3 using different alternative bank interest rate risk measures described in Section 3.3. Columns 1, 3, and 4 use value-weighted bank stock returns as the dependent variable, while column 2 uses equal-weighted returns. "Rate sensitivity" is the measured interest rate exposure, based either on DeMarzo et al. (2024), Begenau et al. (2015), or Haddad and Sraer (2020). "Rate Shock" is the first principal component of 30-minute changes in yields around FOMC announcements, scaled to a unit change in the 5-year Treasury yield (Section 3), using data from Gürkaynak et al. (2022) and Jarocinski (2024). Panel (b) excludes 2007 H2–2009 H1. All standard errors are heteroskedasticity-robust.

Portfolio	**Vol (annualized)**	**Avg log return**	**Max drawdown**
Bank stocks alone	24.9%	12.0%	45.2%
Bank stocks + 50% LT treasuries	22.0%	16.3%	40.7%
Bank stocks + 100% LT treasuries	21.3%	20.2%	39.9%

Table 6: **Volatility of bank stock and Treasury portfolios, 2010–2021.** Each row calculates the volatility, average log return, and maximum drawdown of different portfolios from daily data 2010–2021. Portfolios consist of a unit long position in bank stocks plus a long position in long-term Treasuries, funded at the short-term risk-free rate and sized proportional to the bank stock holdings. Bank stock returns are from the Fama–French bank industry index. Long-term Treasury returns are from the iShares 20+ Year Treasury Bond ETF. Risk-free rates are taken from the Ken French data library.

	Unhedged	Duration-hedged
Asset duration	2.0	5.5
Change in going-concern value from a 1 ppt rise in rates	+0.038	0
Change in liquidation value from a 1 ppt rise in rates	−0.022	−0.060
Rise in rates that exhausts book equity in liquidation (ppt)	4.5	1.7

Table 7: **Duration hedging and liquidation value in a stylized bank.** The bank has \$1 of deposits, \$1.1 of assets and \$0.1 of book equity. Deposit spread sensitivity (0.06 per dollar of deposits per ppt) is from Figure 6 panel (a) near its 2021 trough. Going-concern value includes the value of deposit spreads. Liquidation value is assets minus the face value of deposits. All changes are first-order approximations.

Appendices

A A model of discrete depositor choice

The motivation for this model is primarily practical and empirical, but it can also be microfounded with a simple model of consumer choice. In this section, I derive the model from a setup where depositors make a discrete choice among banks with different quality levels. The only departure from a standard "logit" model as in McFadden (1973) that is needed to match the empirical model from Section 1.1 is that consumers pay more attention to prices when rates are falling than when they are rising.

I focus in this model on depositors' choice of bank, rather than their choice of deposit quantity. This focus is justified if depositors' choice of bank is much more elastic to individual bank deposit rates than their choice of overall savings quantities. Put simply, I assume that when banks set their deposit rates they are primarily worried about customers moving to a different bank, not customers saving less in total.

Suppose there are N depositors indexed by j and M banks indexed by i, where N and M are both large. Each depositor chooses a single bank. The depositor's utility from each bank depends on the deposit pricing (r_{it}^d), but also on the quality of the services provided by that particular bank (ξ_{it}) and some random error term for that bank–customer combination (ε_{jit}) that follows an extreme value distribution. If the bank pays less than 0 interest, the depositor will choose to hold cash instead of deposits, but still retain the benefit of the bank relationship.

The depositor therefore chooses bank i to solve:

$$\max_i U_{jt} = \max\{r_{it}^d, 0\} + m_t \left(\xi_{it} + \varepsilon_{jit}\right)$$

The variance of the quality and error terms is scaled by a potentially time-varying weight m_t, which can be interpreted as inversely related to how much attention the depositor pays

to price or how hard she searches.

Following standard results, the share of deposits demanded from bank i is a logit function of prices:

$$D_{it} = \frac{\exp\left(r_{it}^d/m_t + \xi_{it}\right)}{\sum_{k=1}^{M} \exp\left(r_{kt}^d/m_t + \xi_{kt}\right)}$$

Each bank therefore chooses deposit rates to maximize its total profit relative to funding at the risk-free rate:

$$\max_{r_{it}^d} D_{it}\left(r_t^f - r_{it}^d\right)$$

The bank's first-order condition, after some rearranging, is:

$$r_{it}^d - r_t^f = \frac{-m_t}{1 - D_{it}}$$

Since M is large, each bank's share of deposits (D_{it}) is small, so the equilibrium deposit rate is approximately:

$$r_{it}^d \approx r_t^f - m_t$$

If m_t is constant, banks will have fixed deposit spreads. In reality, as documented in Section 1, we tend to observe wider spreads when rates have recently risen. This effect suggests that bank market power, and thus m_t, is greater when rates have recently risen.

I therefore assume depositors have some lagging perception of the true level of risk-free rate, gleaned from social networks, news, etc., given by:

$$\hat{r}_t = \gamma \hat{r}_{t-1} + (1 - \gamma)\,\beta r_t^f + \varepsilon_t$$

When the true risk-free rate is below the perceived rate (i.e., m_t is low), the depositor thinks she is getting a bad deal and so searches harder and puts more emphasis on price.

Hence the bank market power m_t is given by:

$$m_t = \beta\delta + r_t^f - \hat{r}_t$$

This effect could also be seen as a reduced form for a more complicated model with search costs and frictions where customers are more likely to look into switching banks when they are notified of a rate cut on their account.

It may seem counterintuitive to suggest depositors focus more on price when rates have fallen, since most of us have been more likely to check our bank rates in the past 2 years when rates have increased. But this is a function of (1) the zero floor on deposit rates, which pinned spreads near 0 in an exogenous fashion when rates were low, and (2) the equilibrium pricing behaviors of banks. Banks respond to the lower price-attentiveness of customers in rising rate periods by increasing spreads until depositor choices remain the same.

With this form for m_t, the equilibrium bank deposit rate becomes:

$$\max\{r_{it}^d, 0\}$$

where

$$r_{it}^d \approx \gamma r_{it-1}^d + \beta\left(1-\gamma\right)\left(r_t^f - \delta\right)$$

which matches Equations (1) and (2) for an individual depositor.

The parameters (β, γ, δ) here are characteristics of the depositors, not the bank, and thus all banks charge the same rate in this model. To think about different banks with different parameters, as I estimate in Section 1.1, we should consider multiple product and regional markets with different customer parameters. Each bank has a different exposure to different markets, so faces different parameters. Different product markets also deliver a range of δ values within a bank, which allows for the "phased-in" censoring from Equation (2).

B Comparison of deposit interest rate function estimates to DeMarzo et al. (2024)

This appendix compares my estimates of how deposit rates respond to interest rates with those of DeMarzo et al. (2024). There are three main findings:

1. I can closely replicate the aggregate deposit estimates of DeMarzo et al. (2024) (Table 8).

2. The largest source of the difference between our estimates is the scope of deposits. DeMarzo et al. (2024) study the interest expense on all deposits, while I study the rate on interest-paying instant-access deposits. Changing the deposit scope accounts for about half of the difference in the estimated fixed spread δ, and is also the largest single contributor to the difference in long-term pass-through β. Allowing for the zero lower bound and using my sample period and risk-free rate account for most of the rest (Table 9).

3. Even on the DeMarzo et al. (2024) deposit scope, allowing for the zero lower bound implies large swings in deposit duration (Figure 13). The degree of non-linearity looks small, but bank leverage and the importance of long-horizon cash flows when rates are low turn it into large changes in bank interest rate sensitivity.

B.1 Replicating DeMarzo et al. (2024)

DeMarzo et al. (2024) regress the aggregate deposit spread, as a share of bank assets, on the fed funds rate and on two variables that capture slow adjustment of deposit rates: the average one- and five-year Treasury yields over the previous one and five years, each minus the fed funds rate. Their sample runs from 1984 to 2021. I rebuild their aggregate series from bank-level Call Report data, following their sample definition: commercial banks only, excluding banks with mostly foreign-office deposits, banks with large credit card businesses,

and the smallest banks.[18] Table 8 shows that my estimates are within 0.01 of theirs, with an R^2 of 94.9% compared with their 95.3%.

B.2 Sources of the differences

Table 9 starts from the replication of DeMarzo et al. (2024) in column 1 and changes one element of the specification at a time, until column 7 reproduces my main estimates. All columns report the fixed spread δ and long-term pass-through β of deposit rates, so they are directly comparable to each other and to my main estimates.[19]

The steps are:

- *Replication* (column 1): a fixed spread of only 0.35 ppt and a pass-through of 0.68.
- *Zero lower bound* (column 2): adding my censored model of the zero lower bound raises the fixed spread to 0.64 ppt and the pass-through to 0.71. The fit is almost unchanged.
- *Risk-free rate and sample period* (columns 3 and 4): using the 3-month Treasury yield instead of the fed funds rate, and my sample period of 1987–2023 instead of 1984–2021, raises the fixed spread to 1.2 ppt and the pass-through to 0.80.
- *Bank sample* (column 5): moving to the banks in my sample makes little difference.
- *Deposit scope* (column 6): moving to interest-paying instant-access deposits is the largest single change. The fixed spread rises to 2.7 ppt and the pass-through to 1.0.

[18] Two of these choices matter most. Savings banks reported income semi-annually in the 1980s, which creates large spurious swings in their quarterly deposit expense, so including them lowers the fit of the regression. Banks with mostly foreign-office deposits pay rates that track foreign rather than US rates, so including them lowers the coefficient on the fed funds rate from 0.24 to 0.21. The small remaining differences are likely due to DeMarzo et al. (2024) dividing by tangible rather than total assets, which my data do not allow, and to data cleaning.

[19] DeMarzo et al. (2024) measure the deposit spread per dollar of assets, so in column 1 the pass-through is one minus the coefficient on the fed funds rate in my replication of their regression (Table 8, column 2) divided by the average ratio of deposits to assets: $1 - 0.235/0.74 = 0.68$. Columns 2–5 also model deposit costs per dollar of assets and are converted to rates in the same way.

- *Adjustment for lags* (column 7): replacing the lagged Treasury yield variables with my AR(1) pass-through delay gives my main estimates, with a fixed spread of 1.95 ppt and a pass-through of 0.94. It also improves the fit, reducing the mean absolute error from 0.15 to 0.10 ppt.

B.3 Duration implied by the DeMarzo et al. (2024) deposit scope

The zero lower bound changes the estimates in column 2 only modestly, but it still generates large swings in deposit duration. Figure 13 plots the contribution of deposit value to bank interest rate sensitivity (the negative of duration) implied by the main estimates in panel (a) and by column 2 in panel (b). The column 2 estimates imply a contribution that rises to about 38 in mid-2020 and falls to about −13 by April 2022, a swing of about 50. This is similar to the range of the main estimates, from −11 to 40. With my risk-free rate and sample period (column 4, panel (c)), the DeMarzo et al. (2024) deposit scope implies even larger swings than my main estimates, peaking at about 51 in mid-2020.

A small degree of non-linearity in deposit costs leads to large variation in duration for two reasons. First, leverage: deposits are typically five to six times bank market equity, and more than eight times in 2020, so a small change in the value of a dollar of deposits is a large change in the value of equity. Second, when rates are very low, the value of the deposit franchise depends heavily on spreads at long horizons, so even a modest floor on deposit rates today changes the sensitivity of a long stream of cash flows.

B.4 Choice of deposit scope

I choose to focus on interest-paying instant-access deposits to ensure I am not mixing secular trends in deposit types with the effects of interest rates, particularly for non-interest-paying deposits. The share of domestic deposits that paid no interest saw a steady decline throughout the 1990s and 2000s from a peak of 23% in the late 1980s to 8% just before the financial crisis in 2007. This change mechanically increased aggregate deposit rates as interest rates fell. For example, in 1989 interest-paying instant-access deposit rates were 30 basis points above the aggregate deposit rate, but by 2007 they were 60 basis points

	(1)		(2)	
	Replication	DKN	Replication	DKN
r_t^*	0.255	0.257	0.235	0.241
	(13.12)	(12.97)	(16.99)	(18.41)
ℓ_t^1			-0.231	-0.238
			(-4.30)	(-4.66)
ℓ_t^5			-0.120	-0.122
			(-4.41)	(-4.68)
Intercept	-0.002	-0.002	0.002	0.002
	(-1.57)	(-1.56)	(2.02)	(2.27)
R^2 (%)	79.49	79.16	94.93	95.28
Obs.	150	150	150	150

Table 8: **Replication of DeMarzo et al. (2024) Table 3, Panel A, columns 1 and 2.** Time-series regressions of the aggregate deposit spread per dollar of assets on the fed funds rate (r_t^*) and, in column 2, the one- and five-year term swap ladder payments (ℓ_t^1, ℓ_t^5), 1984 Q1–2021 Q2. "Replication" columns use Call Report data for commercial banks, excluding bank-quarters with a majority of deposits in foreign offices, more than 30% of interest income from credit cards, or assets in the bottom percentile. "DKN" columns are the estimates published in DeMarzo et al. (2024), which divide by tangible rather than total assets. t-statistics in parentheses are based on Newey–West standard errors with 8 lags.

below it.

If we value a static deposit quantity based on this historical aggregate deposit rate, we would be implicitly assuming that if rates increase to their early 1990s levels, the share of non-interest-paying deposits will *increase*. This assumption seems implausible, since we would expect that a higher rate environment will lead depositors to be less willing to hold non-interest-paying deposits.

	(1) As in DKN (2025)	(2) With ZLB censoring	(3) Treasury risk-free rates	(4) Same dates	(5) Same dates & bank sample	(6) Same dates, bank sample & deposits	(7) Main specification
δ Fixed spread	0.004	0.006	0.007	0.012	0.013	0.027	0.019
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.005)
β Long-term pass-through	0.684	0.715	0.731	0.804	0.866	1.012	0.940
	(0.019)	(0.026)	(0.027)	(0.027)	(0.032)	(0.050)	(0.110)
γ Pass-through delay							0.788
							(0.031)
θ_{1y} Loading on 1y swap spread	0.311	0.328	0.292	0.414	0.411	0.267	
	(0.072)	(0.073)	(0.081)	(0.068)	(0.067)	(0.088)	
θ_{5y} Loading on 5y swap spread	0.162	0.176	0.195	0.254	0.290	0.377	
	(0.037)	(0.035)	(0.035)	(0.038)	(0.036)	(0.033)	
s Zero bound phase-in rate		0.007	0.007	0.010	0.012	0.020	0.017
		(0.001)	(0.001)	(0.002)	(0.002)	(0.002)	(0.004)
σ Error standard deviation	0.003	0.003	0.003	0.003	0.002	0.003	0.002
		(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Observations (quarters)	150	150	150	145	145	145	145
R^2	0.994	0.995	0.994	0.994	0.995	0.992	0.997
MAPE (ppt)	0.198	0.194	0.198	0.168	0.158	0.148	0.101
Dependent variable	Deposit cost/assets	Deposit cost/assets	Deposit cost/assets	Deposit cost/assets	Deposit cost/assets	Deposit rate	Deposit rate
Zero-bound censoring	No	Yes	Yes	Yes	Yes	Yes	Yes
Pass-through delay (γ)	No	No	No	No	No	No	Yes
Risk-free rate	Fed funds	Fed funds	3m Treasury	3m Treasury	3m Treasury	3m Treasury	3m Treasury
Deposits	All	All	All	All	All	Int.-paying	Int.-paying
Banks	DKN	DKN	DKN	DKN	Ours	Ours	Ours
Sample start	1984Q1	1984Q1	1984Q1	1987Q2	1987Q2	1987Q2	1987Q2
Sample end	2021Q2	2021Q2	2021Q2	2023Q2	2023Q2	2023Q2	2023Q2

Table 9: **Comparison to DeMarzo et al. (2024) estimation approach.** Each column changes one element of the specification, moving from the replication of DeMarzo et al. (2024) in column 1 to the main estimates of this paper in column 7. In columns 1–5, as in DeMarzo et al. (2024), the dependent variable is interest expense on all deposits per dollar of assets; their estimates are converted to deposit-rate units by dividing by the sample-average ratio of deposits to assets (about 0.74), so that all columns are comparable. In columns 6 and 7 the dependent variable is the interest rate on interest-paying instant-access deposits. "DKN" banks are the DeMarzo et al. (2024) sample of Table 8; "Ours" are the banks used in the main estimates. R^2 is uncentered and MAPE is the mean absolute error of fitted values, in the units of each column's dependent variable. Standard errors in parentheses are Newey–West with 8 lags in column 1 (as in DeMarzo et al. (2024)) and GMM Newey–West with 4 lags in the other columns, as in Table 1.

C Replicating portfolio: long-end assumptions

This appendix shows how the replication-based deposit value and duration of Section 2 depend on implementation choices and assumptions about cap prices beyond the longest quoted maturity. Throughout, the duration measure is the one in Figure 6: the negative of the coefficient from a 250-trading-day rolling regression of daily changes in the value of the replicating portfolio on daily changes in the 10-year zero-coupon yield, either per dollar of deposits or scaled by the ratio of interest-paying instant-access deposits to bank market equity. Summary statistics are calculated over the days with a rolling estimate, mid-2007 to mid-2023.

The baseline assumes that the forward price of the 25–30 year cap segment repeats every five years after 30 years. I consider two alternatives.

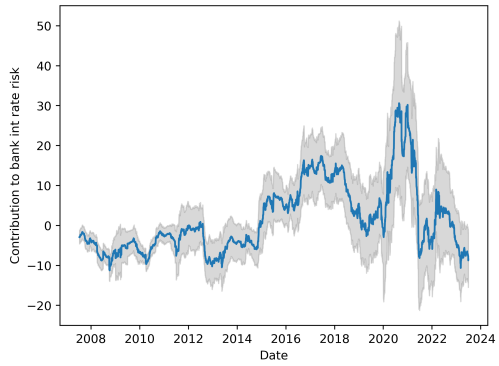
1. *Forward cap price constant after 15 years.* Quoted volumes are lower for long-maturity caps, so their prices may be less reliable. This alternative uses only caps up to 15 years: the forward price of the 10–15 year cap segment is assumed to repeat every five years after 15 years, discounted at the 10–15 year forward rate. The same assumption is applied to the annuity that values the fixed spread paid at zero rates.

2. *Black implied volatility constant after 30 years.* Every caplet beyond 30 years is priced with the Black formula at the implied volatility of the 30-year cap at the same strike, with forward rates and discount factors from the Gürkaynak et al. (2007) curve, whose instantaneous forward rate is held constant after 30 years. Because the variance of the log rate, $\sigma^2 t$, then grows without limit, far-dated caplets approach their forward value. This implies that interest rate uncertainty keeps increasing far into the future, so I view it as an upper bound on the negative duration at low rates.[20]

Figure 14 and Table 10 show the results. The baseline lies between the two alternatives.

[20] A cap is worth at most the floating leg with the same maturity, so the value of the replicating portfolio cannot exceed 1 under this assumption. What changes is the share of value coming from far-dated cash flows, which are the most sensitive to rates.

Panel (a): Deposit contribution to bank interest rate sensitivity (i.e., negative of duration) implied by the main deposit estimates



Panels (b) and (c): Implied by estimates using the DeMarzo et al. (2024) sample and deposit scope, and either (b) the DeMarzo et al. (2024) risk-free rate and method of adjusting for lags (Table 9, column 2), or (c) my risk-free rate and dates (Table 9, column 4)

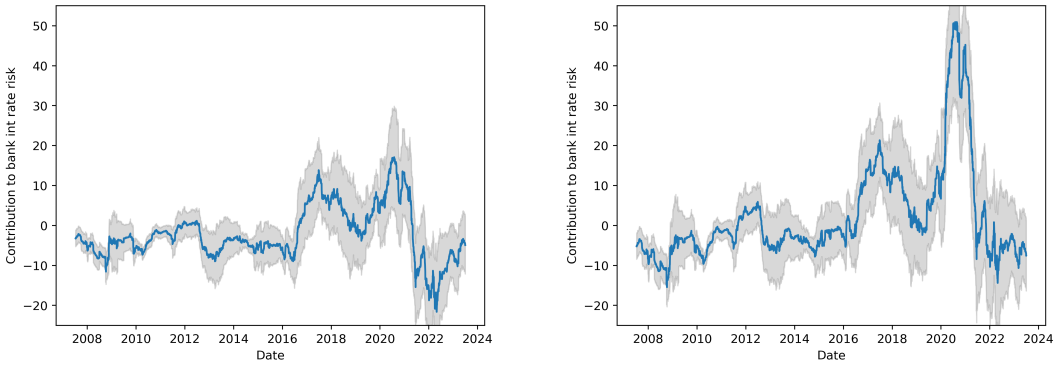


Figure 13: **Deposit effects on bank duration implied by the main body estimates versus estimates similar to DeMarzo et al. (2024).** Each panel plots the deposit contribution to bank interest rate sensitivity implied by the replicating cap portfolio of Section 2, scaled by the ratio of deposits to the market equity of public banks. Panel (a) uses the main estimates and interest-paying instant-access deposits. Panels (b) and (c) use the coefficients from columns 2 and 4 of Table 9, converted to the cost per dollar of deposits by dividing β and s by the average deposits to assets ratio, and all deposits. Shaded areas show 95% confidence intervals.

	Mean	Min	Max	SD
Panel A: Deposit franchise value per dollar of deposits				
Baseline: 25–30y cap segment repeats	0.34	0.09	0.43	0.05
10–15y cap segment repeats	0.39	0.16	0.49	0.04
30y Black vol constant	0.29	0.03	0.37	0.06
Panel B: Rate sensitivity per dollar of deposits (negative of rolling beta)				
Baseline: 25–30y cap segment repeats	1.04	-3.89	8.25	2.64
10–15y cap segment repeats	-0.27	-4.84	4.29	1.89
30y Black vol constant	4.44	-2.04	12.93	3.75
Panel C: Contribution to bank interest rate sensitivity (scaled by deposits / market equity)				
Baseline: 25–30y cap segment repeats	4.6	-10.9	39.8	10.1
10–15y cap segment repeats	0.0	-14.0	23.5	6.8
30y Black vol constant	17.7	-4.7	67.1	16.4

Table 10: **Replicating portfolio value and duration under alternative long-end assumptions: summary statistics.** Statistics over the days with a 250-day rolling estimate, mid-2007 to mid-2023. Specifications are described in Appendix C.

Holding Black volatility constant after 30 years raises the average deposit contribution to bank rate sensitivity from 4.6 to 17.7 and the peak from 40 to 67, and it remains strongly positive in 2021 and 2022. The time pattern is similar to the baseline (correlation 0.83). Relying only on caps up to 15 years gives smaller variation: the standard deviation of the scaled contribution is 6.8, compared with 10.1 in the baseline, and the 2020 peak is about 23 rather than 40. The correlation with the baseline is 0.62. So the most conservative assumption about the long end shrinks the variation in deposit duration by about a third, but it still produces a large positive contribution to rate sensitivity in 2020, and negative contributions in the high-rate periods of 2007–08 and 2023.

D Proof of negative deposit duration at low rates

Denote the value of a continuous call option from time 0 to T given a path of instantaneous forward rates $\{f_t\}_0^\infty$ as:

$$V_C\left(0, T, \{f_t\}_0^T\right) = \int_0^T P(0,t)C(t, f_t)dt$$

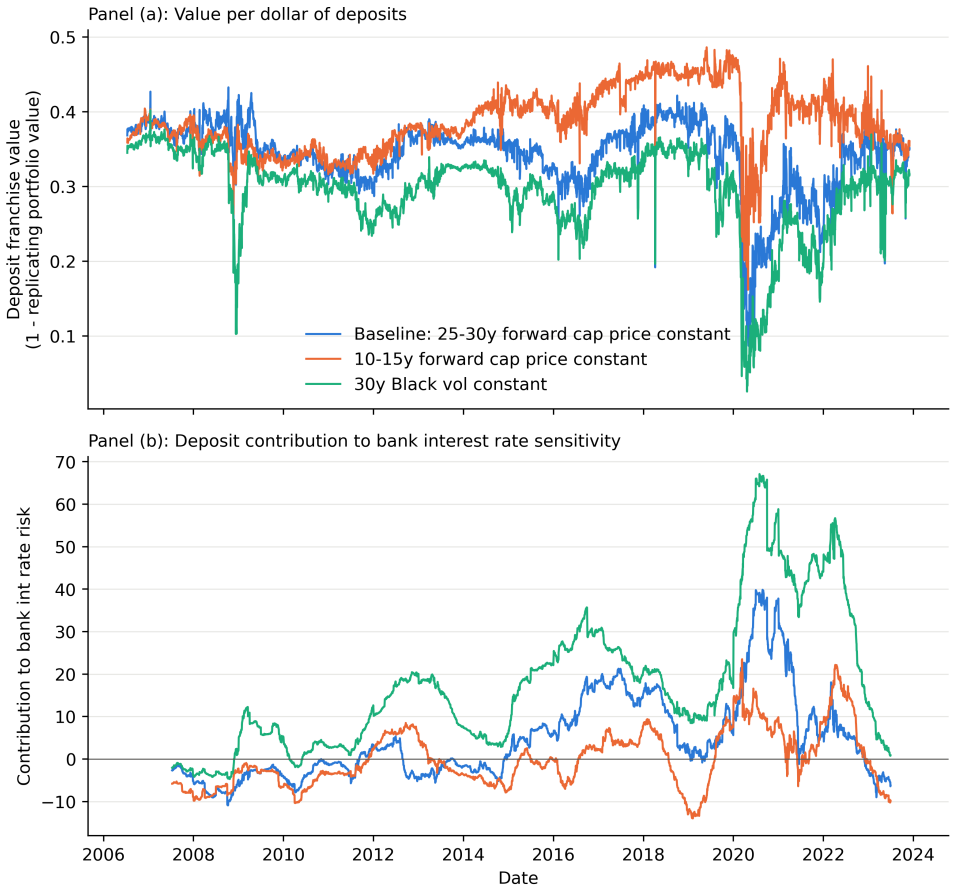


Figure 14: **Replicating portfolio value and duration under alternative long-end assumptions.** Panel (a) plots the value of deposit spreads (1 minus the value of the replicating portfolio) per dollar of interest-paying instant-access deposits. Panel (b) plots the implied deposit contribution to bank interest rate sensitivity, as in Figure 6, Panel (b). "Baseline" assumes the forward price of the 25–30 year cap segment repeats after 30 years. "10–15y forward cap price constant" assumes the forward price of the 10–15 year segment repeats after 15 years. "30y Black vol constant" prices all caplets after 30 years at the 30-year cap's Black implied volatility. All use the continuous replication.

Proposition 1:

Let the path of forward rates $\{f_t\}_0^\infty$ be such that:

- Forward rates are held below some value $\eta > 0$ from time 0 to some time T:

$$\forall t \in [0, T] : \ 0 < f_t < \eta$$

- After T, rates are high enough that the perpetuity rate is at least $\overline{r}$ and the value of the permanent call before discounting is at least $\overline{V}$:

$$\int_T^\infty P(T, s)ds \leq \frac{1}{\overline{r}}$$

$$V\left(T, \infty, \{f_t\}_T^\infty\right) = \int_T^\infty P(T, s)C(s, f_s) \geq \overline{V}$$

And let C satisfy a few simple conditions appropriate for a call price:

- $C(t, f_t) \geq 0$ and $\frac{\partial C}{\partial f_t}(t, f_t) \in [0, 1]$ for all $t \in [0, \infty)$ and $f_t \in (0, \infty)$
- As $f \to 0$, either the call option price is bounded below or it smoothly approaches 0:

$$\forall f > 0, t > 1 : \ C(t, f) > C_0 > 0$$

or

$$\forall t > 0 : \ \lim_{f \to 0} C(t, f) = 0; \lim_{f \to 0} \frac{\partial C}{\partial f_t}(t, f) = 0$$

Then there exists a value of η low enough and T large enough that a positive level shock to forward rates will have an arbitrarily negative effect on the value of the permanent call—i.e., duration is arbitrarily positive:

$$\forall D < 0, \ \exists \eta, T \text{ s.t. } \frac{\partial}{\partial \varepsilon}\Big|_{\varepsilon=0} V_C\left(0, \infty, \{f_t + \varepsilon\}_0^\infty\right) < D$$

where

$$V_C\left(0,\infty,\{f_t+\varepsilon\}_0^\infty\right)=\int_0^\infty P(0,t)C(t,f_t+\varepsilon)dt$$

Proof:

Separate the value of the call by time into three portions, splitting at time 1 and T:

$$V_C\left(0,\infty,\{f_t+\varepsilon\}_0^\infty\right)=v_0^1+v_1^T+v_T^\infty$$

where

$$v_A^B=P(0,A)V_C\left(A,B,\{f_t+\varepsilon\}_A^B\right)$$

And note that the derivative of each with respect to the rate shock (i.e., the negative duration) is given by:

$$\frac{\partial}{\partial\varepsilon}\Big|_{\varepsilon=0}V_A^B=P(0,A)\left(\int_A^B P(A,t)\left(\frac{\partial C}{\partial f_t}(t,f_t)-tC(t,f_t)\right)dt-AV_C\left(A,B,\{f_t\}_A^B\right)\right)$$

I will show the duration of each of the three parts is either bounded or goes to infinity as $\eta\to 0$ and $T\to\infty$. Thus the total duration goes to infinity.

v_0^1: The portion from 0 to 1 must have a derivative ≤ 1 because $\frac{\partial C}{\partial f_t}$ is bounded above at 1 and C is bounded below at 0:

$$\frac{\partial}{\partial\varepsilon}\Big|_{\varepsilon=0}v_0^1\leq\int_0^1 P(0,t)1dt\leq 1$$

v_T^∞: As $\eta\to 0$, the portion after T has a duration that becomes arbitrarily large as T increases, because the value is bounded below at $\overline{V}$:

$$\begin{aligned}\lim_{\eta\to 0}\frac{\partial}{\partial\varepsilon}\Big|_{\varepsilon=0}v_T^\infty&=\int_T^\infty P(T,t)\left(\frac{\partial C}{\partial f_t}(t,f_t)-tC(t,f_t)\right)dt-TV\left(T,\infty,\{f_t\}_T^\infty\right)\\&\leq\int_T^\infty P(T,t)1dt-T\overline{V}\leq\frac{1}{\overline{r}}-T\overline{V}\end{aligned}$$

v_1^T: Depending on the behavior of the call function as $f \to 0$, the duration of the portion between 1 and T will either approach 0 or become arbitrarily large as T gets large.

If the value is bounded below (as in the Vasicek model), then:

$$\begin{aligned}\lim_{\eta\to 0}\frac{\partial}{\partial\varepsilon}\Big|_{\varepsilon=0} v_1^T &= \int_1^T \frac{\partial C}{\partial f_t}(t, f_t) - tC(t, f_t)dt - V\left(1, T, \{f_t\}_1^T\right) \\ &\leq \int_1^T (1 - tC_0)\, dt \leq\ T - 1 - \frac{C_0}{2}(T-1)^2\end{aligned}$$

which is arbitrarily negative as $T \to \infty$.

If the value and derivative of the call option approach 0 as $f \to 0$ (as in a model where rates are lognormal, for example):

$$\lim_{\eta\to 0}\frac{\partial}{\partial\varepsilon}\Big|_{\varepsilon=0} v_1^T = \int_1^T 0dt - 0 = 0$$

Hence as $\eta \to 0$ the derivative of the call option with respect to rates becomes arbitrarily negative as T increases.

E Details of deposit model estimation

E.1 Data filters

For the bank-level analysis I conduct the same estimation separately for each bank. For data quality purposes I drop from the individual analysis:

- Banks with fewer than 50 quarterly observations
- Micro-banks, defined as those with under $100 million in assets in 2023 Q1, or the equivalent percentile of total banking system assets for previous quarters
- Banks with deposits under 1/3 of assets

- Banks that ever report paying deposit rates more than 3 ppt above the fed funds rate, or a 2 ppt jump in rates that reverses the next quarter
- Banks that report negative or exactly 0 rates (just 64 small banks)
- Bank-quarter observations that have a jump of more than 50 bp that reverses the next quarter

E.2 Likelihood function

To estimate the model, I numerically maximize the joint likelihood of the observed deposit rates. The conditional log likelihood of deposit rates can be derived as:

$$L\left(r_{t+1}^{d}\middle|r_{d,t}, r_{t+1}^{f}, \hat{\gamma}, \hat{\beta}, \hat{\delta}, \hat{\sigma}, \hat{s}\right) = \log\left(\frac{1}{\hat{\sigma}}\phi\left(\frac{\hat{\varepsilon}_{t+1}}{\hat{\sigma}}\right)\right) - \log\left(\Phi\left(\frac{f^{-1}\left(r_{t+1}^{d}\right)}{\hat{s}}\right)\right)$$

where

$$\hat{\varepsilon}_{t+1} = f^{-1}\left(r_{t+1}^{d}\right) - \hat{\gamma} f^{-1}\left(r_{t}^{d}\right) - \hat{\beta}\left(1-\hat{\gamma}\right)\left(r_{t+1}^{f} - \hat{\delta}\right)$$

And f^{-1} is the inverse of the censoring function from Equation (2).

The optimization can be sped up by noting that, conditional on s, the maximum likelihood solutions for $(\hat{\gamma}, \hat{\beta}, \hat{\delta}, \hat{\sigma})$ are given by the OLS estimator for Equation (1), plugging in $f^{-1}\left(r_t^d\right)$ for $\widetilde{r}_t$. Hence I can conduct a fast univariate optimization over $\hat{s}$, plugging in the OLS solutions for the other parameters for each guess for $\hat{s}$.

E.3 Bank-level results

Summary statistics for the coefficients estimated from the bank-level analysis for interest-paying instant-access deposits are shown in Table 11. The cross-sectional means are similar to the aggregate estimates, with relatively high values of δ and values of β near one. Fixed spreads (δ) are generally substantial and statistically significant.

The association of δ with bank characteristics, also shown in Table 11, is mostly consistent with market-power and cost-to-serve interpretations of the fixed spreads. High

spreads are correlated with the bank cost base (e.g., from branch costs) and the share of small depositors, measured as the share of deposits under the FDIC insurance thresholds. Surprisingly, high local market concentration is associated with a lower δ.

	Summary stats			Regression coefficients			
	Mean	Stdev	PctSignificant	Log assets	Cost/ assets	Insured share	Cty HHI
δ	0.028	0.017	0.738	-0.179	0.067	0.053	-0.112
γ	0.859	0.080	0.963	-0.835	-0.040	2.243	0.607
s	0.030	0.019	0.789	0.143	0.111	0.299	-0.061
σ	0.003	0.001	0.922	-0.013	0.005	-0.023	-0.002

Table 11: **Summary statistics for bank-level estimates of deposit parameters**. 3,374 banks are included in the savings deposit results and 912 in the transaction deposits. "Share signif" shows the percent that pass a t-test at a 95% confidence level. Parameters and bank characteristics are winsorized at the 1% level. Columns labeled "regression coefficients" show the results of a regression of bank-level average characteristics on the estimated bank-level coefficients. "Insured share" is the share of deposits consisting of balances under the FDIC insurance thresholds ($100 K before 2009 and $250 K after). Characteristics labeled "Cty" are branch-weighted averages of county characteristics. "HHI" is a measure of local market concentration, specifically the Herfindahl–Hirschman index of bank branches. "Age" and "income" are the average age and income of counties from census data.

F Asset duration

A full picture of the interest rate risk of the bank requires us to also calculate asset duration. I do so using Call Report data on maturities and interest income and applying standard bond pricing formulas and mortgage-backed security (MBS) index pricing data. Asset duration varies between 1 and 2 and does not change enough to counteract the swings in deposit duration described in Section 2.

F.1 Methodology & assumptions

Unlike for interest-paying instant-access deposits, I focus on valuing the stock of existing assets rather than the permanent flow of loans. I treat time deposits the same way. I use a different approach for two reasons. First, unlike for instant-access deposits, there is rich information available about the maturity of loans and time deposits. There is no need to try to estimate interest rate exposure from the time series of interest income when we can measure it directly from maturity.

Second, interest rates on assets are above the risk-free rate, not below it. So assets will not show the same spread compression dynamics that are described in Section 1 when interest rates approach 0. Estimation of the duration of expected spreads on new loans is therefore performed in a simple fashion in Section G.

The Call Reports provide maturity data on four types of assets: non-mortgage loans (constituting 43% of assets over 2002–2023), mortgage loans (11%), mortgage-backed securities (11%), and Treasuries (9%). A further 25% of assets have no maturity data, and largely consists of cash and deposits. The average interest income rate is also provided for each of these products.

I estimate the value and duration of non-mortgage loans, Treasuries, and time deposits by assuming they consist of a series of coupon bonds. The Call Reports provide data on the book value of loans maturing in six different maturity buckets (e.g., under 3 months, 3 months to 1 year, etc.). I assume the balance in each bucket matures at the mid-point

of the bucket.

I use the average interest income of the product as the coupon rate. For non-mortgage loans, I apply a credit spread equal to the ICE BofA high-yield bond credit spread index. Duration is generally close to the maturity for coupon bonds, and so is relatively insensitive to coupons or spreads.

Estimating the value and duration of mortgages and MBS is more complicated because of the prepayment option embedded in the product. Fortunately, there are widely available indices used to track the price and duration of mortgage-backed securities (e.g., as used in Hanson, 2014). I apply price and duration estimates from the Bloomberg Barclays MBS indices to the bank mortgage holdings.

F.2 Results

The time series of estimated asset duration and the contribution of each type of product is shown in Figure 15. Unsurprisingly, asset duration is relatively stable over time and does not flip signs like deposit spread duration does.

Short-term trends in duration (e.g., the large increase in 2022–23) are driven by mortgage convexity. When interest rates fall, borrowers tend to refinance their mortgages, lowering the duration of existing mortgages; when rates rise, refinancing slows and duration extends.

There is also a long-term secular increase in asset duration post-crisis. This has two sources. First, banks increased their holdings of long-maturity assets, particularly around 2020. MBS and Treasury holdings rose from around 20% of assets at the start of 2019 to 26% at the end of 2021.

Second, the duration of mortgages and mortgage-backed securities themselves increased from an average of 2 in 2000–2010 to an average of 3.6 in 2013–2023, as adjustable-rate mortgages and shorter-term fixed-rate mortgages have been replaced by 30-year fixed-rate mortgages. The Bloomberg Barclays fixed-rate MBS index increased its average duration

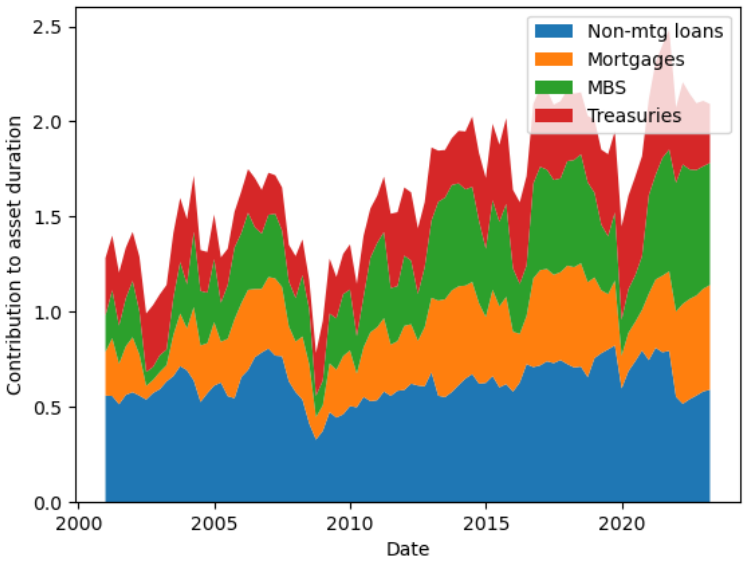


Figure 15: **Aggregate duration of bank assets 2002 Q3–2023 Q2**. The total stacked height shows the aggregate duration of all assets. Each color shows the contribution of a specific product. Duration is defined as $-\frac{\partial V/\partial r}{V}$, where V is value. Details of the estimation methodology are in Appendix F.

from 3 to 4 during this period, and I estimate that the share of adjustable-rate mortgages has fallen from 25% to 6%. These estimates are in line with the Federal Housing Finance Agency's Monthly Interest Rate Survey, which shows ARM shares fluctuating between 20% and 30% pre-crisis and dropping below 10% post-crisis (Moench, Vickery, and Aragon, 2010).

G Total bank duration

The bank does have other expected cash flows besides the deposits and loans: costs, fees, profitability of new loans, any future governmental transfers or levies, etc. A full accounting of bank interest rate risk requires some assumption on the duration of this residual component. In my baseline I assume that the residual cash flows constitute a perpetuity. As an alternative, I assume they have zero duration. This choice leads to a substantial difference in the level of bank interest rate sensitivity, but the time-series patterns are similar (Figure 9).

G.1 Methodology & assumptions

We can check the scale of the residual bank value by comparing my estimates of deposit and asset value to the market value of public banks.

A regression of aggregate bank stock prices on the measured asset and deposit values in Table 12 shows that these two measures alone explain just under half of the variation in bank valuation over the past 20 years.[21]

However, there is a large unexplained portion of bank value. For my main estimation I simply assume that this residual value in each period is composed of a perpetuity c_t. The full detail of this simple calculation as well as the calibrated values are shown in Appendix H.

[21] The regression in levels is only run post-crisis because of the large shift in the level of market-to-book pre- and post-crisis.

=

	Levels (post-crisis)	Annual changes
(Liab value - book) / book equity	0.282***	0.737***
	(0.051)	(0.212)
(Asset value - book) / book equity	0.465***	0.590**
	(0.090)	(0.173)
Intercept	1.541***	−0.099+
	(0.168)	(0.051)
Num.Obs.	58	80
R2	0.495	0.365
Std.Errors	Newey-West (L=4)	Newey-West (L=4)

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Table 12: **Regression of aggregate bank market-to-book on estimated deviations of asset and deposit value from book value, 2002 Q3–2023 Q2**. Regressors are scaled by aggregate book equity. Estimation of asset and deposit franchise value is described in Sections 2 and F. The regression in levels is performed on data from 2009 to 2023 Q2 because of the large level shift in market-to-book around the financial crisis (e.g., on the full sample, a pre-/post-crisis dummy alone has an R^2 of 60%).

The resulting value of aggregate c_t is around -0.2% of assets pre-crisis and around -0.8% post-crisis. The net impact is that estimated bank duration is lower post-crisis. This is reasonable, because the drop in market prices must translate to higher discount rates or lower cash flow expectations, either of which lowers duration.

This approach of choosing a constant long-term cash flow to match equity prices is analogous to the approach used in the implied equity duration literature, specifically Dechow et al. (2004) and Weber (2018). I hold the discount rate constant and vary c_t for numerical simplicity and consistency with the existing literature. If I varied discount rates instead, the results would likely be similar—higher market-to-book would imply higher duration in both cases.

The assumption behind this approach is that the residual cash flows are not a function of interest rates. For operational costs, this assumption is relatively uncontroversial, and is also made, for example, in Drechsler et al. (2023); DeMarzo et al. (2024); Begenau and Stafford (2018).

For spreads on new lending, the evidence points toward only very modest effects of the interest rate level. The spread between rates on new mortgage loans (accounting for 30% of aggregate lending) and MBS yields increases by just 0.05 ppt for each 1 ppt increase in the fed funds rate. These effects are small enough that they could be accounted for by prepayment option effects or borrowers shifting to higher-fee products. DeMarzo et al. (2024) also find that loan spreads have a loading of just 0.05 on the interest rate.[22]

Replacing some of the c_t perpetuity with a floating-rate spread would shift the level of duration, with less change in its time-series pattern. Figure 9 shows the estimated duration if we assume that the residual value comes entirely from floating rates, and the value of c_t is 0.[23]

What would change the overall pattern of duration would be a strong non-linear response of loan spreads to interest rates—for example if loan spreads exactly cancel out the compression of deposit spreads at low interest rates. However, the recent path of net interest margin suggests this is not the case: from 2022 Q1 to 2022 Q4, as rates rose, the spread between bank loan and deposit rates rose by over a full percentage point.[24]

I also show in Appendix H that there is no correlation between changes in the c_t estimates and changes in interest rates. This suggests that I am not systematically over- or under-estimating the residual duration.

H Details of residual value estimation and robustness

I define the future "remaining cash flows" $C_{t,T}$ at time t of each bank such that:

$$P_t = V_{A,t} + V_{DF,t} - V_{L,t} + \int_t^\infty e^{-(T-t)r_{t\to T}} E_t^*(C_{t,T})dT$$

[22] Wang (2020) shows that loan spreads increase as rates fall, but only for two relatively small loan categories: ARMs and vehicle loans.

[23] This is similar to DeMarzo et al. (2024)'s estimation of a nearly 0 fixed franchise perpetuity for the average bank.

[24] The drop in interest rates in 2008–09 was not associated with a drop in net interest margin, most likely due to a large one-off increase in (a) credit spreads and (b) term premia after the financial crisis.

where P_t is the pre-tax market price of bank equity, V_A is the value of assets, V_{DF} the value of the deposit franchise, V_L the book value of liabilities, and $r_{t \to T}$ the spot rate from time t to T.

The pre-tax market price of bank equity is calculated as $P_t = \frac{\text{MktCap}}{1-\tau}$, where MktCap is the market capitalization observed on CRSP and τ is the corporate tax rate, assumed here to be 0.25.

I assume that at time t the expected future remaining cash flows are a constant share c_t of bank assets A_t. Similar to the paper's main deposit valuation estimates, I assume that the risk premium is equal to the expected growth rate and hence do not apply any additional discount rate over the risk-free rate:

$$E^*(C_{t,T}) = A_t c_t$$

I find c_t by matching the value of the observed market price of the bank in each period:

$$c_t = \frac{P_t - V_{A,t} - V_{DF,t} + V_{L,t}}{A_t \int_t^\infty e^{-(T-t)(r_{t \to T})} dT}$$

I test the approach by checking for correlation between changes in c_t estimates and changes in interest rates. If I systematically over-estimated duration, then I would find that the value of the bank is higher than predicted after rates increase. This would mean that there should be a correlation between changes in interest rates and changes in the value of c_t. Table 13 reports the coefficients from a regression of changes in c_t on changes in short- and long-term interest rates. The point estimates are close to 0 and insignificant and the 95% confidence intervals do not include any large effects.

The estimated values of the aggregate rate of expected cash flows c_t for all public banks are shown in Figure 16. The most striking feature of the time series is the drop in expected remaining cash flows around the global financial crisis. Prior to 2007, c_t remained around -0.2% to 0% of assets, whereas from 2010 onwards it was generally around -0.8% or

	Δc_t
Δ Fed funds rate	0.016
	(0.030)
Δ 10y yield	0.008
	(0.059)
Intercept	−0.001*
	(0.000)
Num.Obs.	84
R2	0.015
Std.Errors	Newey-West (L=4)

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Table 13: **Regression of annual changes in aggregate c_t on annual changes in interest rates, 2003 Q3–2023 Q2.** c_t is the estimated residual cash flow perpetuity described in Appendix G. Regressors are changes in the federal funds rate and the yield on 10-year Treasury securities. c_t and yields are both expressed in the same units, so a coefficient of 0.01 implies a 1 ppt increase in rates is associated with a 0.01 ppt increase in c_t.

below.

Summary statistics for the panel of c_t estimates are shown in Table 14. The cross-sectional standard deviation is high, at 1.5 ppt on average over 2003–2023. c_t is only weakly correlated with bank size and costs. Unsurprisingly, since c_t is constructed as the portion of market-to-book not explained by liability and asset valuations, it also shows strong positive correlations with market-to-book and negative correlations with deposit franchise value and asset value.

While c_t plays an important role in determining the level of duration, the time-series patterns are similar whether or not it is included. Figure 9 plots the time series of bank duration under the baseline and under the alternative assumption that the residual cash flows have zero duration. In both cases, interest rate sensitivity peaks when long rates are low and the floor is valuable.

Figure 17 decomposes the baseline measure, showing the contribution of interest-paying

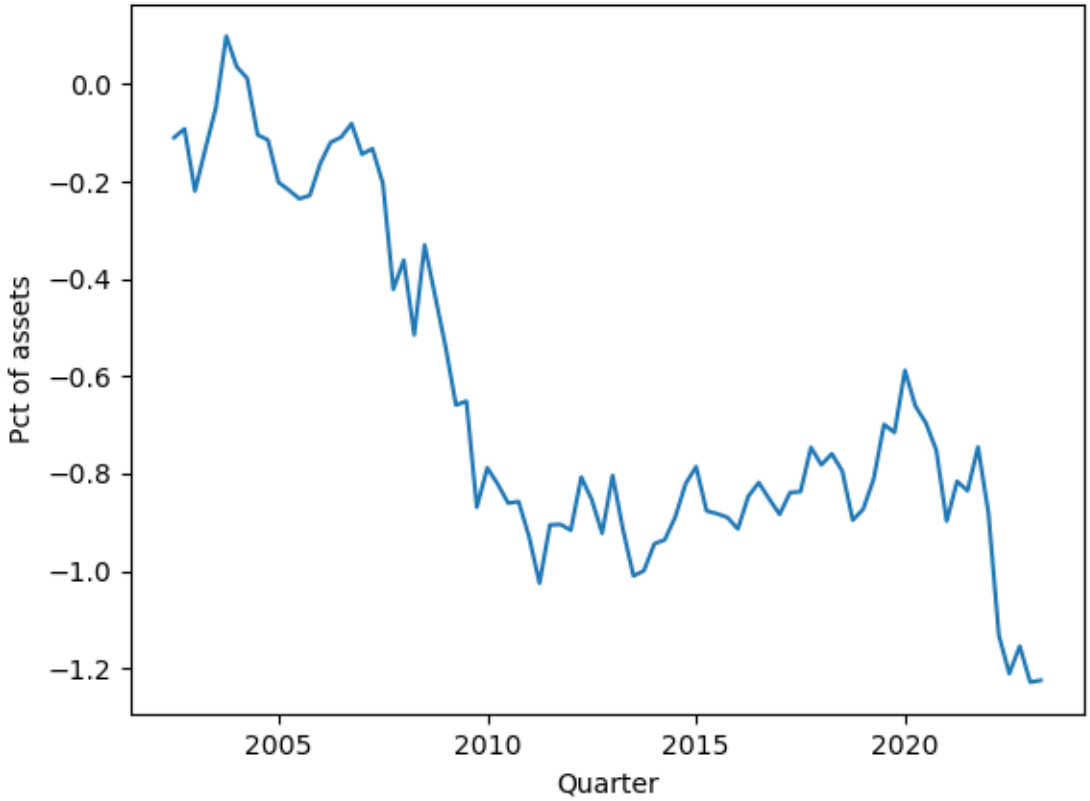


Figure 16: **Fitted values for the residual cash flow expectation perpetuity c_t, 2002 Q3–2023 Q2**. See Appendix G for description of methodology.

Summary stats	Summary stats	Correlations	Correlations
# obs	34604.00	Log assets	0.04
# banks	942.00	Cost/ assets	0.05
Avg obs per qtr	407.11	Maket-to-book	0.57
Mean c_t (ppt)	-4.07	(Liab value - book) / BE	-0.09
Std of c_t (ppt)	21.18	(Asset value - book)/ BE	-0.05
Cross-sectional std c_t	19.28		

Table 14: **Summary statistics for estimates of residual cash flow perpetuities c_t for public banks, 2002 Q3–2023 Q2.** c_t is the estimated residual cash flow perpetuity described in Appendix G. Parameters and bank characteristics are winsorized at the 1% level. Columns labeled "Correlations" show the correlation of parameters with the average across quarters of different bank characteristics.

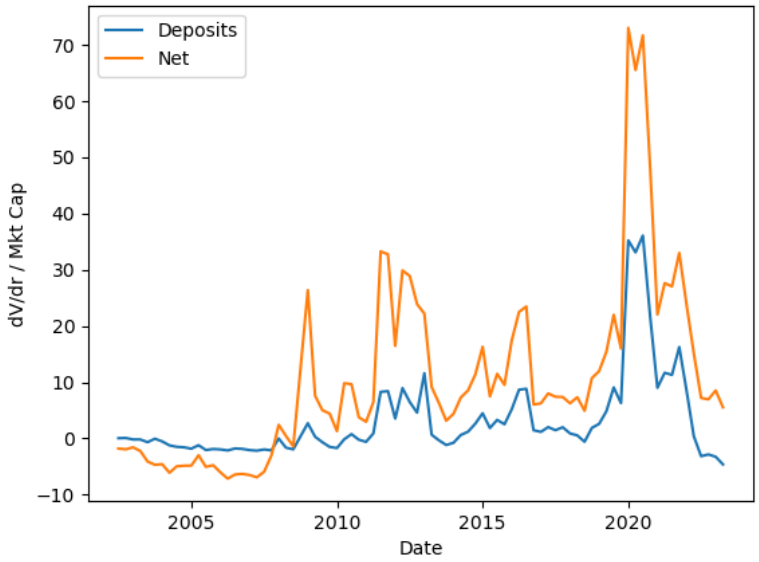


Figure 17: **Aggregate interest rate sensitivity of deposits and of the total bank, scaled by market capitalization, 2002 Q3–2023 Q2**. The blue line plots the estimated derivative of deposit value with respect to a monetary shock, divided by market capitalization (i.e., the deposit contribution to negative duration) of all publicly listed US banks. The orange line plots the total, including assets and the baseline perpetuity for non-asset, non-deposit cash flows.

instant-access deposits alongside the total. The deposit contribution resembles the replication-based estimates in Figure 6: it is positive when rates are low, peaking in 2020, and slightly negative when rates are high.

I Comparison to other sources of variation in aggregate duration

Mortgage-backed security prepayment risk and insurance company convexity and duration mismatch are two key factors often credited for driving investor interest rate exposure and long-term yields (e.g., Hanson, 2014; Carboni and Ellison, 2022; Domanski et al., 2017). Figure 18 plots the aggregate duration created by these two sources over the past 20 years

vs the bank duration described in this paper, and shows that swings in bank duration are larger and more persistent. The standard deviation of total bank duration is approximately 3–4× higher than either MBS or insurer duration.

J Additional stock return validation tests

Figure 19 compares the measured banking sector duration from Appendix G to the coefficients from a simple rolling one-quarter regression of daily bank stock returns on changes in the 10-year yield, with no high-frequency identification.

Bank stock price behavior shows the same patterns and scale as this paper's duration measure. When yields were low and the zero floor on deposit rates was binding, the relationship between yields and bank stock prices became positive.

Table 15 repeats the aggregate high-frequency regressions from Table 3 using duration measures calculated from expanding-window estimates of the deposit parameters, which do not use any data from after the date of each interest rate shock.

K Duration tests under alternative specifications

This appendix examines which parts of the bank valuation drive the high-frequency results in Table 3. For three different deposit valuations, I calculate three measures of aggregate bank interest rate sensitivity: deposits only, the whole bank with the residual cash flows c_t assumed to have zero duration, and the whole bank with c_t valued as a perpetuity, as in the baseline (Appendix G). The residual c_t is recalculated from the market value of bank equity for each deposit valuation, so a higher deposit value implies a lower residual.

The three deposit valuations are:

1. *Main.* The valuation in Section 2: interest-paying instant-access deposits are valued using the estimates from Section 1, non-interest-paying deposits pay no interest, and time deposits are valued as fixed-rate liabilities.

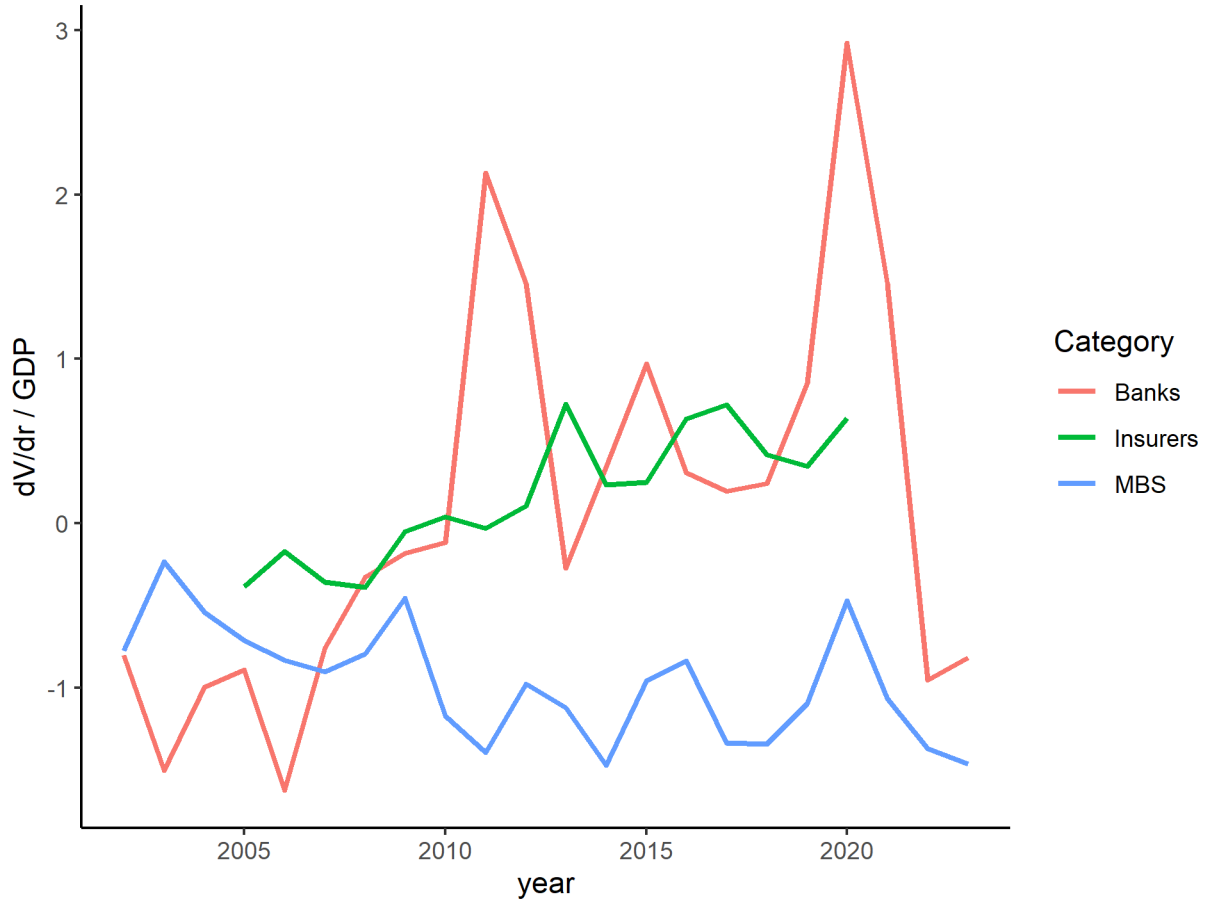


Figure 18: **Comparison of time-series of total dollar duration from the bank mechanism described in this paper, insurers, and mortgage-backed securities, 2002–2023**. The y-axis shows the negative of dollar duration, scaled by GDP. Insurer duration is taken from Huber (2022). MBS duration is provided by the Bloomberg Barclays aggregate MBS index and matches the figures used in Hanson (2014). Bank dollar duration is estimated on publicly listed banks and then scaled up to all banks using the ratio of all bank assets to public bank assets.

	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Rate sens. × rate shock	1.216**	1.089***	0.429*	1.162***	1.197***	0.638***
	(0.377)	(0.282)	(0.170)	(0.330)	(0.341)	(0.185)
Rate shock	−4.802	−7.155	3.759	−1.944	−3.674	3.601+
	(6.851)	(5.160)	(2.466)	(2.637)	(2.957)	(2.165)
Rate sensitivity	0.003	0.003	−0.001	0.005	0.004	0.000
	(0.009)	(0.008)	(0.007)	(0.008)	(0.008)	(0.007)
ΔIG spread		23.747***	12.283***		10.975	10.683
		(5.005)	(2.582)		(17.900)	(9.222)
ΔHY spread		−10.340*	−2.967		−3.615	−2.565
		(5.084)	(1.900)		(3.814)	(1.805)
Market return			1.524***			1.246***
			(0.146)			(0.119)
Intercept	0.185	0.035	−0.147+	0.120	0.115	−0.071
	(0.168)	(0.190)	(0.078)	(0.128)	(0.127)	(0.065)
Num.Obs.	168	168	168	152	152	152
R2	0.062	0.200	0.839	0.104	0.119	0.738

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Table 15: **Out-of-sample aggregate measured duration, calculated using expanding windows, predicts bank stock price response to interest rate shocks.** This table repeats the regressions in Table 3 using bank duration measures calculated from expanding-window estimates of bank interest rate sensitivity that do not use any data from after the date of the relevant interest rate shock. "Rate sensitivity" is the measured interest rate exposure of the aggregate banking system, described in Appendix G, and calculated without any future data. "Rate Shock" is the first principal component of 30-minute changes in yields around FOMC announcements, scaled to a unit change in the 5-year Treasury yield (Section 3), using data from Gürkaynak et al. (2022) and Jarocinski (2024). "Δ IG spread & HY spread" denote controls for daily changes in the ICE BofA option-adjusted investment-grade or high-yield credit spread index. "Market return" is a control for the CRSP total market daily return. Columns 4–6 exclude 2007 H2–2009 H1. All standard errors are heteroskedasticity-robust.

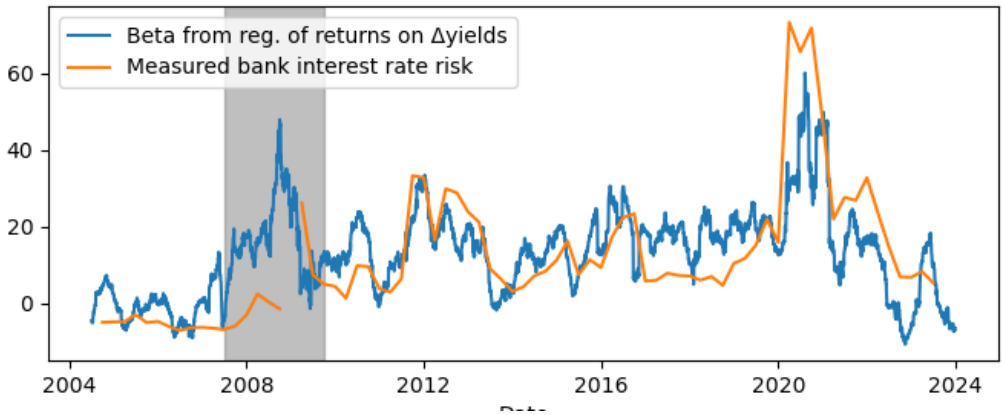


Figure 19: **The bank interest rate risk measure closely tracks the relationship between daily yield changes and daily bank returns.** The orange line shows the aggregate interest rate sensitivity of public banks. Sensitivity is calculated as the negative duration of all bank equity under the baseline assumptions (details in Appendix G). The blue line shows the coefficient from a rolling one-quarter regression of one-day banking sector stock returns on one-day changes in the 10-year Treasury yield. Data from the financial crisis (2007 H2–2009 H1) are shaded in gray.

2. *All deposits, censored model.* All deposits are valued together, using the censored autoregressive model of Section 1.1 estimated on the interest rate on all deposits, the deposit scope of DeMarzo et al. (2024). As in the main specification, β is set to one. The estimates are $\delta = 1.36$ ppt, $s = 1.75$ ppt, and $\gamma = 0.80$, compared with $\delta = 2.16$ ppt, $s = 1.94$ ppt, and $\gamma = 0.79$ for interest-paying instant-access deposits.

3. *All deposits, linear DKN model.* All deposits are valued with the linear model of DeMarzo et al. (2024), in which deposit costs are $\beta\left(r^f - \delta\right)$ with no zero lower bound. I use the parameters from my replication of DeMarzo et al. (2024) in Appendix B (Table 9, column 1): $\beta = 0.68$ per dollar of deposits and $\delta = 0.35$ ppt. Deposit value is then a floating-rate component, $(1 - \beta)$ per dollar of deposits, with no interest rate sensitivity, plus a small fixed perpetuity, $\beta\delta$, with positive duration.

Figure 20 plots the nine measures, and Table 16 repeats the regression of banking sector returns on FOMC announcement days on the interaction of each measure with the rate shock (columns 1 and 4 of Table 3).

The pattern is consistent with the deposit mechanism. Under both censored-model valuations, the deposits-only measure predicts the response of bank stocks to rate shocks, with positive coefficients that are significant at the 5% level in both the full sample and excluding the financial crisis (columns 1 and 4). But the fixed cost perpetuity is also an important driver of the main result. Without it, there would not be enough negative duration: the whole-bank measures with $c_t = 0$ are negative (i.e., banks have positive duration) in most quarters outside 2020–21, and their interaction coefficients are above one and, for the main valuation, insignificant (columns 2 and 5). With the perpetuity included, the coefficients are close to one in both samples and I cannot reject that they are equal to one (columns 3 and 6).

An estimation based on deposits with approximately zero duration, the linear DeMarzo et al. (2024)-style model, underestimates the variation in duration. Under this model, deposits have slightly positive duration, so the deposits-only and $c_t = 0$ measures have the wrong sign (columns 7 and 8), and all of the time variation in whole-bank interest rate sensitivity comes from the residual perpetuity. The whole-bank measure still predicts the response to rate shocks (column 9), but its interaction coefficient is 1.5 in the full sample and 2.2 excluding the financial crisis: the measure varies too little over time to match the observed response of bank stocks. When the financial crisis is excluded, I can reject that the coefficient is equal to one ($p = 0.01$).

As a further test, Table 17 regresses bank stock returns on changes in the long-run discount rate series of Gormsen and Lazarus (2026), which is purged of changes in growth expectations and risk, interacted with the main measure of bank interest rate sensitivity. The series is semi-annual until 2014 and quarterly thereafter, so each observation is the change between consecutive observations of the series, matched with the banking sector return over the same interval and the interest rate sensitivity at the end of the previous quarter. The interaction coefficient is significant at the 5% level, although at 0.57 it is well below one. The coefficient on the change in the discount rate itself, which gives the predicted response of a bank with zero measured interest rate sensitivity, is large (-24) but only marginally significant ($p = 0.08$). It is possible that this reflects some information

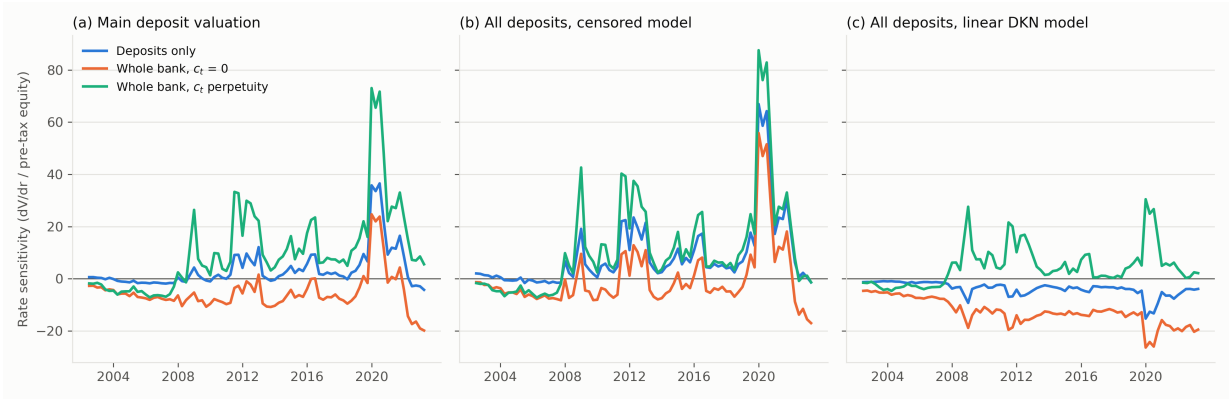


Figure 20: **Aggregate bank interest rate sensitivity under alternative deposit valuations.** Interest rate sensitivity (negative of duration) of aggregate public bank equity, measured as the derivative of value with respect to the interest rate divided by pre-tax market equity, 2002 Q3–2023 Q2. Each panel uses a different deposit valuation, described in Appendix K. "Deposits only" includes only the value of deposits (including time deposits). "Whole bank, $c_t = 0$" adds the assets and assumes the residual cash flows have zero duration. "Whole bank, c_t perpetuity" values the residual cash flows as a perpetuity, as in the baseline. The residual c_t is recalculated for each deposit valuation. 2008 Q4 is omitted.

about bank cash flows or risk that remains in the measure, even though it is residualized from changes in GDP growth expectations and the VIX.

	Main deposit valuation			All deposits, censored model			All deposits, linear DKN model		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Deposits only	$c_t = 0$	Full	Deposits only	$c_t = 0$	Full	Deposits only	$c_t = 0$	Full
Panel A: Full sample									
Rate sens. × rate shock	3.226*	1.457	1.263**	1.823**	1.806*	0.983***	−5.679***	−2.426***	1.489***
	(1.475)	(1.368)	(0.413)	(0.668)	(0.822)	(0.264)	(1.517)	(0.705)	(0.395)
Rate shock	−5.905	8.189	−7.516	−10.699	3.381	−7.516	−18.537*	−27.131*	−5.601
	(6.984)	(8.679)	(7.460)	(8.383)	(5.261)	(7.243)	(7.705)	(10.893)	(6.581)
Rate sensitivity	0.006	0.016	0.003	0.004	0.006	0.004	−0.015	−0.004	0.013
	(0.016)	(0.020)	(0.008)	(0.009)	(0.010)	(0.006)	(0.046)	(0.027)	(0.016)
Intercept	0.191	0.341*	0.181	0.155	0.206	0.167	0.161	0.175	0.173
	(0.165)	(0.166)	(0.179)	(0.178)	(0.146)	(0.170)	(0.223)	(0.350)	(0.159)
p-value: interaction = 1	0.133	0.739	0.525	0.220	0.328	0.948	0.000	0.000	0.218
Num.Obs.	168	168	168	168	168	168	168	168	168
R2	0.050	0.014	0.061	0.060	0.046	0.057	0.044	0.038	0.043
Panel B: Financial crisis excluded									
Rate sens. × rate shock	2.625*	1.083	1.158***	1.593**	1.531*	1.055***	−7.458***	−2.190**	2.156***
	(1.120)	(1.106)	(0.311)	(0.560)	(0.705)	(0.286)	(1.904)	(0.671)	(0.457)
Rate shock	−3.568	7.858	−4.732*	−7.258*	4.977	−4.512+	−19.431***	−21.552***	−2.361
	(3.398)	(9.043)	(2.370)	(3.414)	(4.357)	(2.390)	(4.724)	(5.640)	(2.449)
Rate sensitivity	0.013	0.024	0.005	0.007	0.010	0.005	−0.020	−0.003	0.014
	(0.015)	(0.019)	(0.007)	(0.009)	(0.010)	(0.006)	(0.042)	(0.026)	(0.015)
Intercept	0.096	0.296+	0.105	0.076	0.145	0.100	0.080	0.132	0.120
	(0.126)	(0.160)	(0.129)	(0.132)	(0.117)	(0.130)	(0.177)	(0.293)	(0.132)
p-value: interaction = 1	0.149	0.940	0.612	0.292	0.453	0.849	0.000	0.000	0.012
Num.Obs.	152	152	152	152	152	152	152	152	152
R2	0.082	0.031	0.103	0.097	0.074	0.111	0.112	0.064	0.103

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Table 16: **High-frequency duration tests under alternative deposit valuations.** This table repeats the regressions in columns 1 and 4 of Table 3 using the interest rate sensitivity measures in Figure 20. The dependent variable is the banking sector stock return on FOMC announcement days, 2002 Q3–2023 Q2. "Rate sensitivity" is the measured interest rate exposure of aggregate public bank equity under each deposit valuation (columns 1–3: main; 4–6: all deposits, censored model; 7–9: all deposits, linear DeMarzo et al. (2024) model) and each treatment of the residual cash flows, described in Appendix K. "Rate Shock" is the first principal component of 30-minute changes in yields around FOMC announcements, scaled to a unit change in the 5-year Treasury yield (Section 3), using data from Gürkaynak et al. (2022) and Jarocinski (2024). Panel A excludes duration calculated using 2008 Q4, and Panel B excludes 2007 H2–2009 H1. "p-value: interaction = 1" tests that the coefficient on the interaction term equals one, the value implied by a correctly scaled interest rate sensitivity measure. All standard errors are Newey–West with 3 lags.

L Bank stock and Treasury portfolios after 2021

Table 18 repeats the exercise in Table 6 over 2010–2025 (Panel A) and over 2022–2025 (Panel B). After rates rose in 2022, adding long-term Treasuries to a portfolio of bank stocks increased its volatility and maximum drawdown, rather than reducing them. Over the full 2010–2025 period, the volatility reduction from adding long-term Treasuries is smaller than over 2010–2021, and the maximum drawdown of the portfolio with the largest Treasury position is substantially larger than that of bank stocks alone.

Panel A: 2010–2025

Portfolio	Vol (annualized)	Avg log return	Max drawdown
Bank stocks alone	24.4%	12.1%	45.2%
Bank stocks + 50% LT Treasuries	22.5%	13.8%	45.1%
Bank stocks + 100% LT Treasuries	23.2%	14.9%	59.0%

Panel B: 2022–2025

Portfolio	Vol (annualized)	Avg log return	Max drawdown
Bank stocks alone	22.6%	12.5%	29.8%
Bank stocks + 50% LT Treasuries	24.1%	6.0%	45.0%
Bank stocks + 100% LT Treasuries	28.0%	1.2%	58.4%

Table 18: **Volatility of bank stock and Treasury portfolios, 2010–2025 and 2022–2025.** This table repeats Table 6 for different sample periods. Each row calculates the volatility, average log return, and maximum drawdown of different portfolios from daily data. Portfolios consist of a unit long position in bank stocks plus a long position in long-term Treasuries, funded at the short-term risk-free rate and sized proportional to the bank stock holdings. Bank stock returns are from the Fama–French bank industry index. Long-term Treasury returns are from the iShares 20+ Year Treasury Bond ETF. Risk-free rates are taken from the Ken French data library.

	Bank return
Rate sens. $\times$ $\Delta\rho^*$	0.568*
	(0.214)
$\Delta\rho^*$	−24.264+
	(13.770)
Rate sensitivity	0.472***
	(0.122)
6-month interval	8.904*
	(3.575)
Intercept	−5.961*
	(2.370)
Num.Obs.	60
R2	0.280

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Table 17: **Measured duration and the response of bank stocks to changes in long-run discount rates.** This table shows the results of a regression of banking sector stock returns on changes in the long-run discount rate ρ^* from Gormsen and Lazarus (2026), interacted with the baseline measure of bank interest rate sensitivity, 2002 Q3–2023 Q2. ρ^* is observed semi-annually (Q1 and Q3) until 2014 Q1 and quarterly thereafter. Each observation is the change in ρ^* between consecutive observations, in percentage points, and the value-weighted banking sector return over the same interval, in percent. "Rate sensitivity" is the measured interest rate exposure of aggregate public bank equity at the end of the quarter before each interval, described in Appendix G. "6-month interval" is an indicator for the semi-annual observations. Standard errors are Newey–West with 2 lags.